Nonlinear Conjugate Gradient with Perturbed Heavy-Ball Saddle Escape: Complexity and Local Convergence

Benjamin Séguin Clément W. Royer

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Abstract

We propose and analyze a first-order method for smooth nonconvex optimization that combines a nonlinear conjugate gradient (NCG) iteration, safeguarded by the restart condition of [10] with a backtracking line search, and a perturbed heavy-ball (HB) iteration. While the gradient is large the method takes safeguarded NCG steps; this is the descent phase. When the gradient falls below a tolerance ε_g the iterate becomes a candidate second-order stationary point (SOSP): the method perturbs it, runs a fixed number of HB steps at constant momentum $\gamma < \frac{1}{2}$ with the safeguard suspended, and tests the resulting decrease in f; this is the escape phase. A sufficient decrease certifies a strict saddle and returns control to the descent phase; its absence certifies, with high probability, an $(\varepsilon_g, \varepsilon_H)$-SOSP. Under Lipschitz continuity of the gradient and the Hessian alone, the method returns such a point in $O(\varepsilon_g^{-2} + \varepsilon_g^{-\max\{1+p,\, 2(1+p-q)\}}) + \widetilde{O}(L_1 L_2^2 \Delta_f\, \varepsilon_H^{-4})$ iterations, the escape phase requiring gradient evaluations only and no negative-curvature oracle. The first term is the gradient descent rate and is paid only when the safeguard forces a restart; the second is paid on conjugate steps and its exponent, which the restart parameters set, can be brought below 2. Under the coupling $\varepsilon_H = \sqrt{L_2 \varepsilon_g}$ the escape term dominates and the bound becomes $\widetilde{O}(\varepsilon_g^{-2})$, matching perturbed gradient descent [27]; the accelerated variant [28] attains $\widetilde{O}(\varepsilon_g^{-7/4})$. Under the strict saddle property, a local phase converges to a local minimizer at a rate set by the same exponents. Numerical experiments on two strict-saddle instances compare the method against both, in iterations and in oracle evaluations.

Contents

Chapter 1

Introduction

Nonconvex optimization is the backbone of training machine learning models and many data science tasks can be modeled as nonconvex problems. Typical examples include matrix completion [24], robust PCA [39] and phase retrieval [4]. In this work, we consider the unconstrained optimization problem

$$\min_{x \in \mathbb{R}^n} f(x), \tag{1.1}$$

where $f : \mathbb{R}^n \to \mathbb{R}$ is a twice Lipschitz continuously differentiable nonconvex function. We are interested in finding an $(\varepsilon_g, \varepsilon_H)$-second-order stationary point (SOSP), that is, a point x such that

$$\|\nabla f(x)\| \leq \varepsilon_g \quad \text{and} \quad \lambda_{\min}(\nabla^2 f(x)) \geq -\varepsilon_H, \tag{1.2}$$

where $\|.\|$ denotes the Euclidean norm in $\mathbb{R}^n$. Much of the literature couples the two tolerances by setting $\varepsilon_H = \sqrt{L_2 \varepsilon_g}$, which reduces (1.2) to a single parameter; we quote such results in terms of ε_g and state throughout which convention is in force. The two-tolerance form is more general, and Theorem 1 is stated in it.

Our goal is to build a first-order algorithm for finding such a point and to analyze its complexity. Finding a first-order stationary point (FOSP) is not enough in the nonconvex setting, since such a point may be a local maximum or a saddle. Second-order methods reach an $(\varepsilon_g, \varepsilon_H)$-SOSP with strong complexity guarantees, but require Hessian information that is often out of reach in high dimension. A line of work initiated by [27] showed that perturbed first-order methods reach an SOSP with high probability and with explicit complexity, using gradients alone. No such guarantee is available for nonlinear conjugate gradient (NCG) methods, and obtaining one is the object of this work: we pair an NCG descent phase with a separate saddle escape mechanism built on the heavy-ball (HB) iteration, for which the perturbation argument does close.

1.1 Contributions

The main contribution of this work is a second-order complexity guarantee for a method whose descent phase is an NCG iteration. Theorem 1 shows that PRNCG-HB returns an $(\varepsilon_g, \varepsilon_H)$-second-order stationary point with probability at least $1 - \delta_{\mathrm{tot}}$, under gradient- and Hessian-Lipschitz continuity alone, after at most

$$\underbrace{\frac{\Delta_f}{c_{\mathcal{R}}} \varepsilon_g^{-2}}_{\text{restart steps}} + \underbrace{\frac{\Delta_f}{c_{\mathcal{N}}} \varepsilon_g^{-\max\{1+p,\, 2(1+p-q)\}}}_{\text{conjugate steps}} + \underbrace{\widetilde{O}\Big(\frac{L_1 L_2^2 \Delta_f}{\varepsilon_H^4}\Big)}_{\text{escape windows}}$$

iterations. The first two terms count the descent phase and the third the escape phase. Within the descent phase the count splits again, according to whether the safeguard on the conjugate direction intervenes: the ε_g^{-2} term, which is the gradient-descent rate, is charged only to steps where restart happens, and the second term to steps where the conjugate direction is kept. The exponent of that second term is a free parameter of the safeguard and can be made strictly smaller than 2, it is $\frac{3}{2}$ for the values used in Chapter 4, where the restart never fires. The method is therefore not bounded below by the gradient-descent rate on its descent phase, though the worst case retains the first term.

The distinction disappears under the single-tolerance coupling $\varepsilon_H = \sqrt{L_2\varepsilon_g}$, where the escape term is already $\widetilde{O}(L_1\Delta_f\,\varepsilon_g^{-2})$ and dominates the descent phase whatever the exponents. That bound matches perturbed gradient descent [27] in both the ε_g- and the L_1-dependence, and falls short of the $\widetilde{O}(\varepsilon_g^{-7/4})$ of its accelerated counterpart [28]. Stating Theorem 1 with two tolerances is what makes the descent exponent visible.

The escape analysis follows the perturbation technique of [27, 28]: two trajectories launched from nearby points separate geometrically, so that the set of perturbations from which the method fails to escape is thin in the direction of most negative curvature. Adapting it to a HB window requires work that is not needed for gradient descent, since the linear recursion governing the difference of the two runs is driven by a non-symmetric matrix that does not diagonalize in the eigenbasis of the Hessian. The same structure forces the window to run at constant stepsize and momentum, with the safeguard suspended. The two phases are otherwise independent: the escape guarantee uses nothing from the descent phase beyond the small-gradient trigger, so any scheme providing a per-step decrease could occupy it.

Chapter 4 reports experiments on two strict-saddle instances. Measured in iterations the comparison with PAGD [28] is mixed; measured in oracle evaluations, where an accelerated step costs four to our one, the method leads on both instances. The experiments also show that the constants prescribed by the analysis are far too conservative to be executed directly.

Finally, under the additional assumption that f satisfies the strict saddle property, Chapter 5 shows that the certified point lies in a strongly convex basin, from which a local phase converges to a local minimizer at a rate set by the same exponents.

1.2 Organisation

The rest of this work is organized as follows. Chapter 2 surveys the related work. Chapter 3 presents the algorithm and the main complexity result (Section 3.1), then proves it: Section 3.2 analyses the descent phase, Section 3.3 the escape phase, the heart of this work, and Section 3.4 combines the two into the proof of Theorem 1. Chapter 4 reports the numerical experiments, and Chapter 5 adds a local convergence statement under an additional landscape assumption. Chapter 6 concludes and discusses future work.

1.3 Notations

Throughout this work, we denote $g_k \coloneqq \nabla f(x_k)$ and $s_k \coloneqq x_{k+1} - x_k$. We reserve the global index k for the main iteration counter of Algorithm 1. Within an escape window we use a local index $t \in \{0, \dots, T\}$ and write $\hat{x}_t$ for the corresponding iterate, the two being related by $\hat{x}_t = x_{k_{\text{pert}}+t}$ (Section 3.1.2); quantities derived from a window iterate, such as g_t, s_t and d_t, carry the same local index. We write $B_x(R) \coloneqq \{y : \|y - x\| \le R\}$ for the closed ball of centre x and radius R in $\mathbb{R}^n$; when a ball in another dimension is needed, we indicate it by a superscript, as in $B_x^{(n-1)}(R)$. We use $\mathcal{O}$, Ω and Θ in the usual sense of upper, lower and two-sided bounds up to constants, and $\tilde{\mathcal{O}}$ to hide polylogarithmic dependence.

Chapter 2

Literature Review

2.1 The strict saddle landscape

Finding a global minimizer of a nonconvex function is intractable in general, so one settles for stationarity conditions; and first-order stationarity is too weak, since it does not distinguish a minimizer from a saddle. The $(\varepsilon_g, \varepsilon_H)$-SOSP of (1.2) is the standard target. Higher-order conditions have also been considered: [2] give an efficient algorithm converging to a third-order local optimum, which escapes degenerate saddles that the first two derivatives cannot distinguish from local minima, but show that finding a fourth-order local optimum is NP-hard. Such conditions require access to higher derivatives, and we restrict attention throughout to (1.2).

For many problems of interest an SOSP is not simply a stationary point: for tensor decomposition [18], dictionary learning [53], phase retrieval [54], matrix sensing [3] and completion [19], and some neural networks [29], every local minimizer is global and every saddle is strict, so that reaching an SOSP is close to reaching a global minimizer. Saddle points are then the sole obstruction, and they are abundant: [15] argue that in high dimension critical points are overwhelmingly saddles rather than local minima. More generally, the strict saddle property of [18] asks that every point be either far from stationarity, or at a saddle with a direction of sufficiently negative curvature, or already close to a local minimizer in whose neighbourhood f is strongly convex. Under that property, an $(\varepsilon_g, \varepsilon_H)$-SOSP with small enough tolerances is guaranteed to lie near a local minimizer. When in addition every local minimizer is global, as in the problems above, it lies near a global one.

The property has also been exploited inside algorithms rather than only in their analysis: on Riemannian manifolds, [21] show that a trust-region method applied to strict saddle functions reaches an approximate local minimizer in a number of iterations depending only logarithmically on the accuracy parameter, an improvement over the bounds available for general nonconvex functions (Section 2.2). We return to a Euclidean method exploiting the same structure in Section 2.3.

2.2 Second-order methods

Methods that use curvature information reach an $(\varepsilon_g, \varepsilon_H)$-SOSP with strong complexity guarantees. Newton's method is not among them: near a saddle the Hessian is indefinite and the Newton direction is attracted to every critical point, minimizers and saddles alike. What is needed is a mechanism that exploits negative curvature rather than merely tolerating it.

Trust-region methods [11] provide such a mechanism. At each iteration they minimize the quadratic model

$$m_k(d) = f(x_k) + \nabla f(x_k)^\top d + \tfrac{1}{2} d^\top \nabla^2 f(x_k) d$$

over a ball $\|d\| \le \Delta_k$, adjusting Δ_k according to the agreement between model and function. When $\nabla^2 f(x_k)$ is indefinite the exact solution of this subproblem lies on the boundary, along the most negative eigendirection, so negative curvature is exploited automatically. Classical second-order convergent trust-region schemes reach a point satisfying (1.2) in at most $O(\max\{\varepsilon_g^{-2}\varepsilon_H^{-1},\, \varepsilon_H^{-3}\})$ iterations [9].

Cubic regularization [38] replaces the ball constraint by a cubic penalty, minimizing

$$f(x_k) + \nabla f(x_k)^\top d + \tfrac{1}{2} d^\top \nabla^2 f(x_k) d + \tfrac{\sigma_k}{6}\|d\|^3.$$

Under Assumption 2 and for $\sigma_k \ge L_2$ this model majorizes f, so its global minimizer yields a genuine decrease. In its basic form this improves the trust-region bound to $O(\max\{\varepsilon_g^{-2}, \varepsilon_H^{-3}\})$, the difference being attributable to the restriction the trust-region constraint places on the step norm [9]. The best known bound for a method using second-order derivatives and Newton-type steps, $O(\max\{\varepsilon_g^{-3/2}, \varepsilon_H^{-3}\})$, was established for cubic regularization of Newton's method in [38] and is attained by the adaptive cubic regularization framework of [7], which removes the need to know L_2. Under the coupling $\varepsilon_g = \varepsilon_H = \varepsilon$ all these bounds reduce to $O(\varepsilon^{-3})$, an order that is sharp for the class of second-order methods [9]; taking instead $\varepsilon_H = \varepsilon^{1/2}$ and $\varepsilon_g = \varepsilon$ yields $O(\varepsilon^{-3/2})$, again optimal within that class [8].

The cost of these guarantees is Hessian access. Forming $\nabla^2 f(x_k)$ requires $O(n^2)$ storage, and solving the subproblem through a factorization costs $O(n^3)$ operations per iteration [40], which is prohibitive in high dimension. Trust-region and cubic-regularization can instead be implemented with inexact subproblem solves that access the Hessian only through matrix-vector products: the truncated conjugate gradient method [52] for the trust-region subproblem, and Lanczos iterations [30] for the smallest eigenvalue. Each such product can be computed in $O(n)$ time and space, without forming the Hessian [46], so the per-iteration cost becomes a number of products rather than a factorization. Pushing further, Neon [59] and Neon2 [1] extract an approximate negative-curvature direction from gradient evaluations alone, removing Hessian access entirely.

Curvature information can therefore be obtained cheaply. What remains is the question of how often it must be obtained, and how it combines with the first-order steps that surround it. These are the subject of the next section.

2.3 Hybrid first-order and negative-curvature methods

Curvature information is cheap per query, but a second-order method queries it at every iteration. Hybrid methods reduce the number of queries by separating two regimes. While the gradient is large, they take inexpensive first-order steps; when it falls below ε_g, the iterate is a candidate SOSP and the method invokes a curvature oracle. If a direction d with $d^\top \nabla^2 f(x) d \le -\varepsilon_H \|d\|^2$ is found, moving along it escapes the candidate; if none exists, the point satisfies (1.2) and the method stops. A textbook presentation of this template, combining gradient descent with negative-curvature descent, is given in Section 9.3.1 of [57].

The complexity argument is a budget count over the two step types. Under Assumption 1, a first-order step with $\|\nabla f(x)\| > \varepsilon_g$ decreases f by $\Omega(\varepsilon_g^2)$; under Assumption 2, a step along a direction of curvature at most $-\varepsilon_H$ decreases it by $\Omega(\varepsilon_H^3)$. Since the total decrease is bounded by Δ_f, the number of steps of each type is bounded separately, and the resulting complexity is the sum of the two counts. This is the accounting that this work follows in Section 3.4, with the escape phase in place of the curvature oracle.

Several algorithms realize this template. [50] propose a line-search method in which each iteration computes a search direction, then follows it with a backtracking line search. The direction is Newton-like when the Hessian is positive definite, and a negative-curvature direction otherwise. This yields second-order complexity guarantees for a purely line-search scheme. [51] replace the exact linear algebra by the conjugate gradient and Lanczos methods, so that the

Hessian is accessed only through matrix-vector products, and [13] obtain the analogous result in a trust-region framework.

A parallel line combines negative curvature with acceleration rather than with plain descent, and obtains a better dependence on ε. [5] develop a variant of accelerated gradient descent which either converges at the rate it would enjoy on a convex function, or produces a certificate that the function is nonconvex along the path taken; exploiting that certificate yields a deterministic, dimension-free method reaching $\|\nabla f(x)\| \leq \varepsilon_g$ in $O(\varepsilon_g^{-7/4}\log(1/\varepsilon_g))$ gradient and function evaluations. [6] attain the same $O(\varepsilon_g^{-7/4}\log(1/\varepsilon_g))$ complexity with an accelerated method that additionally guarantees $\nabla^2 f(x) \succeq -O(\varepsilon_g^{1/2})I$ at the returned point, and which is Hessian-free: only gradient computations are required.

Curvature information is what these methods still pay for. Whether it is extracted through Hessian-vector products, as in the Newton-CG and trust-region schemes above, or from gradient evaluations alone, as in the accelerated methods and in Neon, it must be obtained at every candidate stationary point. In its cheapest Hessian-vector form, each such query costs $\min\{n, O(\varepsilon_H^{-1/2})\}$ products, up to a factor logarithmic in ε_H and in the failure probability, to either certify $\nabla^2 f(x_k) \succeq -\varepsilon_H I$ or return a direction of curvature below $-\varepsilon_H/2$ [51]. That cost is not incidental: [42] note that verifying the second-order condition this way would destroy the logarithmic dependence they obtain, and lower-bound the spectrum through ∇f instead.

2.4 Deterministic avoidance of saddle points

Since second-order methods are expensive, it is natural to ask whether curvature information is needed at all to avoid saddle points, or merely useful. A line of work addresses this by casting an algorithm as a dynamical system $x_{k+1} = g(x_k)$ and applying the Center-Stable Manifold Theorem: when g is a local diffeomorphism, the set of initializations converging to a strict saddle is contained in a lower-dimensional manifold and therefore has Lebesgue measure zero. [32] first used this for gradient descent with a small constant step size, and [31] extended it to a wide variety of first-order methods, including manifold gradient descent, block coordinate descent and mirror descent, showing that neither second-order information nor randomness beyond the initialization is needed to avoid strict saddles.

These results have since been strengthened in two directions. On the hypotheses about f, [44] remove the requirement that critical points be isolated and allow non-globally-Lipschitz smoothness on forward-invariant convex subspaces. On the hypotheses about the algorithm, the theorem in its classical form applies only to *autonomous* systems, in which the same map is used at every iteration; this forces a constant step size and excludes the schedules used in practice. [45] prove a stable manifold theorem for time non-homogeneous systems and obtain avoidance for gradient, mirror, manifold and proximal point descent with vanishing step sizes. [35] treat a modified Armijo backtracking method with arbitrary initial step size, dispensing with the Lipschitz gradient assumption, and [36] establish a general Center-Stable Set Theorem for non-autonomous systems $x_{k+1} = g_k(x_k)$ covering vanishing step sizes without assuming Lipschitz gradients or isolated saddles. Parallel results exist in the Riemannian setting, for retraction-based stochastic methods [26] and for non-isolated saddle sets under projected gradient descent [25].

Two limitations remain. The conclusions are asymptotic and carry no complexity guarantee: [16] exhibit natural random initialization schemes and non-pathological functions on which gradient descent takes exponential time to escape, and show that perturbed gradient descent does not suffer this slowdown. And the argument itself is unavailable to the method of this work. Even in the non-autonomous setting the maps g_k must be local diffeomorphisms of a common functional form; the nonlinear conjugate gradient recursion changes form at every restart, and its coefficient β_k depends nonlinearly on the current and previous gradients. This is one reason this work proceeds by perturbation rather than by dynamical-systems arguments.

2.5 Perturbed first-order methods

The results of Section 2.4 guarantee that saddle points are avoided, but not how fast. To obtain complexity guarantees while keeping to first-order oracles, a second line of work adds randomness to the iteration itself rather than only to the initialization.

[18] first showed that a stochastic gradient method with an isotropic perturbation added at every iteration converges to a point close to a local minimizer in polynomially many iterations, under the assumption that the objective satisfies the strict saddle property (Definition 1); the dependence on the dimension is polynomial, and is hidden in their constants. [27] removed the landscape assumption and sharpened the rate: their perturbed gradient descent (PGD) adds a uniform perturbation only when the gradient is small, and reaches an $(\varepsilon_g, \varepsilon_H)$-SOSP of an arbitrary smooth nonconvex objective in a number of iterations that depends only polylogarithmically on the dimension, matching the $\widetilde{O}(\varepsilon_g^{-2})$ rate of gradient descent to first-order stationarity up to logarithmic factors. The analysis rests on a characterization of the geometry around a saddle: the set of perturbations from which the method fails to escape is thin in the direction of most negative curvature, so a uniformly drawn perturbation misses it with high probability. This is the argument this work adapts in Section 3.3.

[28] extend the construction to Nesterov's accelerated gradient descent, obtaining an SOSP in $\widetilde{O}(\varepsilon_g^{-7/4})$ iterations, the first single-loop accelerated method provably faster than gradient descent in the general nonconvex setting. Their analysis introduces the *improve or localize* principle: along a stretch of iterations, either the objective decreases substantially or the iterates remain confined to a small ball, which we use in Lemma 5.

The perturbation need not be large. [34] analyze gradient descent in over-parameterized models and show that infinitesimal perturbations suffice to escape saddle points while keeping the iterates in an implicit low-dimensional region, which yields guarantees for over-parameterized matrix sensing.

Analogous results hold on manifolds. [56] obtain an ε_g^{-2} rate for perturbed Riemannian gradient descent, and [12] extend the analysis of [27] with a distinction between steps taken on the manifold and perturbed steps taken in a tangent space, reaching an SOSP in $O((\log n)^4 \varepsilon_g^{-2})$ gradient queries and so recovering the near-dimension-free behaviour of the Euclidean case.

All of these methods take gradient descent, or a variant of it, as the underlying iteration. This work asks what happens when the underlying iteration is an NCG method instead.

2.6 Momentum in nonconvex optimization

The escape phase of this work uses a HB iteration, so we review what is known about momentum methods outside the convex setting.

The HB method [48] was introduced for convex problems, where its guarantees are classical. On nonconvex functions the objective is no longer monotonically decreasing along the iterates: the momentum term can carry a step uphill, so the descent argument that underlies gradient methods is unavailable. Convergence is instead established through a Lyapunov function combining the objective with the squared displacement, as in [60] and in the iPiano analysis [41]. The energy of Section 3.3.1 is of this form, and $\gamma < \frac{1}{2}$ is the condition under which it decreases monotonically at the fixed stepsize $\alpha_{\mathrm{w}} = 1/L_1$ used inside the escape window.

Momentum also changes the behaviour near saddle points. By the stable manifold argument of Section 2.4, [43] show that the HB method converges to a strict saddle only from a set of initializations of measure zero, and that on a nonconvex quadratic both heavy ball and accelerated gradient move away from such a point faster than steepest descent. [55] obtain the same avoidance result under a weaker stepsize restriction: in the notation of this work, their analysis permits $\alpha_{\mathrm{w}} < 2(1-\gamma)/L_1$, against the $\alpha_{\mathrm{w}} < 1/L_1$ required for gradient descent in [31], a weaker restriction whenever $\gamma < \frac{1}{2}$. Their construction raises the two difficulties we meet in Section 3.3: an iteration of HB involves the two previous iterates, so it is not of the form $x_{k+1} = g(x_k)$ and they work instead with a map on $\mathbb{R}^{2n}$ acting on the pair (x_k, x_{k-1}), whose differential is non-symmetric. The difference state of (3.16) and the matrix $\mathbf{M}$ of (3.17) are the counterparts of these two objects.

Restarts have been combined with momentum to sharpen complexity. [33] run classical accelerated gradient or HB iterations until the accumulated displacement $k\sum_{t<k}\|x_{t+1} - x_t\|^2$ exceeds a threshold, at which point the momentum is reset to zero; both variants reach an ε_g-approximate FOSP in $O(\varepsilon_g^{-7/4})$ gradient evaluations, without negative-curvature exploitation and without the $O(\log \varepsilon_g^{-1})$ factor carried by [28]. Perturbing the restart point when the gradient there is small turns the accelerated variant into a second-order method, at the cost of restoring the logarithmic factor: it reaches a point satisfying (1.2) with $\varepsilon_H = \sqrt{L_2\varepsilon_g}$ in $O(\varepsilon_g^{-7/4}\log(n/\zeta))$ gradient evaluations with probability $1-\zeta$. The restart also lets them take $\alpha_{\mathrm{w}} = O(1/L_1)$ with γ close to one, where the analyses above would require $\alpha_{\mathrm{w}} = O((1-\gamma)/L_1)$; the escape window of Section 3.3 runs at $\gamma < \frac{1}{2}$, where that restriction is not binding.

The descent phase of the method is a different matter. A nonlinear conjugate gradient step also augments the gradient with a multiple of the previous direction, but the coefficient is computed from the gradients rather than held fixed, which places it outside the scope of the results of Section 2.4 and is why the escape mechanism of this work is perturbation-based.

2.7 Nonlinear conjugate gradient

The linear conjugate gradient method was introduced in 1952 for solving positive definite linear systems. Its extension to general smooth objectives [17] replaces the exact stepsize by a line search and the residual by the gradient, giving the two-term recurrence (3.2). The appeal is that it retains the low per-iteration cost and memory footprint of gradient descent, storing only one additional vector, while typically converging much faster; it remains a standard choice for large-scale smooth problems where second-order information is out of reach.

Individual methods differ by the choice of the coefficient β_k, and a substantial literature is devoted to it. The Fletcher–Reeves rule [17] converges globally under strong Wolfe conditions but is prone to generating poor search directions and long sequences of tiny steps; the Polak–Ribière–Polyak (PRP) rule [47] behaves better in practice but can fail to converge, which [20] correct by truncating it at zero (PRP+). The survey of [22] covers the intervening variants.

Complexity guarantees for these methods are more recent. [10] equip NCG with a restart condition (3.5): a direction is accepted only if it is sufficiently downhill and of controlled norm relative to the gradient, and the method falls back on a gradient step otherwise. This safeguard yields a worst-case complexity for reaching an ε_g-FOSP comparable to that of gradient descent, while leaving the conjugate directions untouched whenever they are well behaved. Their descent phase is the one we use in Section 3.2, and Lemmas 1 and 2 are their decrease guarantees.

No complexity guarantee for reaching a *second*-order stationary point is available for any NCG method, and closing that gap is the object of this work. The two routes surveyed above are both obstructed. The dynamical-systems argument of Section 2.4 does not apply, for the reasons given there. The perturbation argument of Section 2.5 does not apply either, at least not directly: it compares two trajectories launched from nearby points, and the comparison closes only if the two runs use the same stepsize and the same momentum coefficient, which a line search and a nonlinear β_k do not guarantee (Remark 6). We therefore do not analyze NCG in isolation. Instead we retain it where it is effective, in the descent phase, and hand control to a separate escape mechanism at candidate stationary points, which is where the perturbation argument does close.

Chapter 3

Nonlinear Conjugate Gradient and Perturbed Heavy Ball

3.1 Setting, Algorithm, and Main Result

Throughout this work, we assume that the three following assumptions hold.

Assumption 1. $f\colon \mathbb{R}^n \to \mathbb{R}$ is twice continuously differentiable and its gradient is L_1-Lipschitz: $\|\nabla f(x) - \nabla f(y)\| \le L_1 \|x - y\|$ for all x, y.

Assumption 2. $\nabla^2 f$ is L_2-Lipschitz: $\|\nabla^2 f(x) - \nabla^2 f(y)\| \le L_2 \|x - y\|$ for all x, y.

Assumption 3. f is bounded below by f_{low}, and $\Delta_f \coloneqq f(x_0) - f_{\text{low}} < \infty$.

The algorithm we propose treats the problem of finding a SOSP by segmenting the optimization process in two distinct phases: a descent phase, and an escape phase, following the structure of [27, 28, 34]. In the descent phase, a safeguarded NCG method is used, introduced in [10]. We run the safeguarded NCG until we reach a low gradient point, which becomes a candidate SOSP, and we shift from the descent to the escape phase. From this point, we run T HB steps on a perturbed iterate and compare the resulting function value against the snapshot $\tilde{x}$. We say the escape *succeeds* when

$$f(\tilde{x}) - f(\hat{x}_T) \;\ge\; F_{\text{cert}}, \tag{3.1}$$

that is, when the T HB steps decrease f by at least the certified amount F_{cert}. If (3.1) holds, we conclude that $\tilde{x}$ is a strict saddle point, discard it, and return to the descent phase to seek a new candidate. If (3.1) fails, the perturbation did not produce enough decrease, and we conclude, with high probability, that $\tilde{x}$ is an $(\varepsilon_g, \varepsilon_H)$-second-order stationary point and return it. We repeat this process until a SOSP is returned.

The main algorithm is therefore a combination of safeguarded NCG and perturbed HB steps. In this section we present both optimization schemes individually and show how to combine them to form the main algorithm that we propose. We also display our main theorem, which is then proven in the rest of this work.

3.1.1 Restarted Nonlinear Conjugate Gradient

NCG methods [22, 49] generate iterates through a two-term recurrence,

$$x_{k+1} = x_k + \alpha_k d_k, \qquad d_{k+1} = -g_{k+1} + \beta_{k+1} d_k, \tag{3.2}$$

where d_k is the search direction, $d_0 = -g_0$, and α_k is a stepsize obtained by Armijo backtracking line search [14]. The update (3.2) has the same structure as a momentum step, the previous direction playing the role of the momentum term; the two families differ in how the coefficient β_{k+1} is set.

Individual NCG methods are distinguished precisely by that choice [17, 47, 23]. We work with the two classical formulas, Fletcher–Reeves,

$$\beta_{k+1}^{\mathrm{FR}} := \frac{\|g_{k+1}\|^2}{\|g_k\|^2}, \tag{3.3}$$

and the truncated Polak–Ribière–Polyak variant PRP+, due to [47] in its original form and to [20] in the truncated one used here,

$$\beta_{k+1}^{\mathrm{PRP+}} := \max\left\{\frac{g_{k+1}^\top(g_{k+1}-g_k)}{\|g_k\|^2}, 0\right\}. \tag{3.4}$$

The method we use is safeguarded by a restart condition from [10]: given $\sigma \in (0,1]$, $\kappa \geq 1$, $p \geq 0$ and $q \geq 0$, the direction d_{k+1} is accepted only if

$$g_{k+1}^\top d_{k+1} \leq -\sigma\|g_{k+1}\|^{1+p} \quad \text{and} \quad \|d_{k+1}\| \leq \kappa\|g_{k+1}\|^q. \tag{3.5}$$

If either condition fails, the method restarts: it discards d_{k+1} and takes a gradient descent step, $d_{k+1} = -g_{k+1}$. The rationale is that the first condition keeps the angle between the search direction and the gradient bounded away from orthogonality, while the second prevents the direction's norm from drifting too far from the gradient's; when either degrades, the method recalibrates to steepest descent. The restart condition also yields a per-step decrease guarantee on the function values, as shown in Section 3.2.

3.1.2 Perturbed Heavy-Ball

The HB method [48] is a classical momentum method; like NCG [58], it augments the gradient step with a term along the previous displacement. Its classical guarantees are convex in nature, an accelerated rate on strongly convex quadratics and accelerated local convergence near a strongly convex minimizer, but not Nesterov's accelerated rate [37] on general smooth convex functions. Its behaviour on *nonconvex* landscapes, and near saddle points in particular, is more subtle, and is what we analyze here: we use the momentum term not to accelerate descent but to drive escape from strict saddles.

Once the iterate reaches a candidate SOSP, we perturb it and run HB from the perturbed point. Formally, when $\|g_k\| \leq \varepsilon_g$ at some step k_{pert}, we freeze the current iterate as a snapshot $\tilde{x} := x_{k_{\mathrm{pert}}}$ and draw $\xi \sim \mathrm{Unif}(B_0(R))$, where R is the perturbation radius. Writing $\hat{x}_0 := \tilde{x} + \xi$ for the perturbed starting point, the escape phase runs T HB steps $\hat{x}_0, \ldots, \hat{x}_T$: for $\gamma \in (0, \frac{1}{2})$ and $\alpha_{\mathrm{w}} = 1/L_1$,

$$\hat{x}_{t+1} = \hat{x}_t - \alpha_{\mathrm{w}}\, g_t + \gamma\,(\hat{x}_t - \hat{x}_{t-1}), \qquad t = 0, \ldots, T-1, \tag{3.6}$$

with $\hat{x}_{-1} := \hat{x}_0$, so that the window starts from rest.

The window-local index is the one introduced in Section 1.3, with $\hat{x}_t = x_{k_{\mathrm{pert}}+t}$ on the global timeline. Two features of (3.6) distinguish the escape phase from the descent phase. The stepsize and the momentum coefficient are held constant, with no line search; the reason is structural and is given in Remark 6. The restriction $\gamma < \frac{1}{2}$ is what makes the HB energy of Section 3.3.1 monotone. Equivalently, and as implemented in Algorithm 1, the window can be written in direction form,

$$d_{t+1} = -g_{t+1} + \gamma\, d_t, \qquad \hat{x}_{t+1} = \hat{x}_t + \alpha_{\mathrm{w}}\, d_t, \qquad d_0 = -g_0.$$

Algorithm 1 PRNCG-HB: Perturbed restarted NCG with HB escape

Input: $x_0 \in \mathbb{R}^n$; L_1, L_2; tolerances $\varepsilon_g, \varepsilon_H > 0$; Armijo $\eta \in (0,1)$, backtracking $\theta \in (0,1)$; restart parameters $\sigma \in (0,1]$, $\kappa \geq 1$, $p \geq 0$, $q \in [0, 1+p]$; momentum $\gamma \in (0, \frac{1}{2})$; failure probability $\delta_{\text{tot}} \in (0,1)$; gap bound Δ_f; escape window stepsize $\alpha_{\text{w}} \leftarrow 1/L_1$.
Derived constants (Section 3.4): window length T, perturbation radius R, target decrease F, and slack $\omega \in (0,1)$; set certified decrease $F_{\text{cert}} \leftarrow (1-\omega)F$.
$g_0 \leftarrow \nabla f(x_0)$; $d_0 \leftarrow -g_0$
for $k = 0, 1, 2, \ldots$ **do**
  if $\|g_k\| \leq \varepsilon_g$ **then** $\triangleright$ candidate SOSP: perturb and run an escape window
    $\tilde{x} \leftarrow x_k$
    $\hat{x}_0 \leftarrow \tilde{x} + \xi$, $\xi \sim \text{Unif}(B_0(R))$
    $\hat{x}_{-1} \leftarrow \hat{x}_0$ $\triangleright$ start from rest
    for $t = 0, 1, \ldots, T-1$ **do** $\triangleright$ HB
      $\hat{x}_{t+1} \leftarrow \hat{x}_t - \alpha_{\text{w}} \nabla f(\hat{x}_t) + \gamma(\hat{x}_t - \hat{x}_{t-1})$
    end for
    if $f(\tilde{x}) - f(\hat{x}_T) \geq F_{\text{cert}}$ **then** $\triangleright$ escape succeeded: $\tilde{x}$ is a strict saddle (w.h.p.)
      $x_{k+1} \leftarrow \hat{x}_T$
      $g_{k+1} \leftarrow \nabla f(\hat{x}_T)$
      $d_{k+1} \leftarrow -g_{k+1}$ $\triangleright$ hand back to NCG, fresh direction
    else $\triangleright$ escape failed: $\tilde{x}$ is an $(\varepsilon_g, \varepsilon_H)$-SOSP (w.h.p.)
      return $\tilde{x}$ $\triangleright$ Theorem 1
    end if
  else $\triangleright$ descent phase: safeguarded NCG with backtracking
    $\alpha_k \leftarrow \theta^{j_k}$, j_k minimal with $f(x_k + \theta^{j_k} d_k) \leq f(x_k) + \eta\, \theta^{j_k} g_k^\top d_k$
    $x_{k+1} \leftarrow x_k + \alpha_k d_k$
    $g_{k+1} \leftarrow \nabla f(x_{k+1})$
    $d_{k+1}^{\text{trial}} \leftarrow -g_{k+1} + \beta_{k+1} d_k$
    if (3.5) holds for d_{k+1}^{trial} **then**
      $d_{k+1} \leftarrow d_{k+1}^{\text{trial}}$
    else
      $d_{k+1} \leftarrow -g_{k+1}$
    end if
  end if
end for

3.1.3 The algorithm

Merging the two schemes of Sections 3.1.1 and 3.1.2 yields Algorithm 1.

Each pass of the outer loop is either a descent step or a full escape window, according to whether the trigger $\|g_k\| \leq \varepsilon_g$ fires. A window consumes T inner iterations and returns either a certified SOSP, in which case the method stops, or the window endpoint $\hat{x}_T$, from which the descent phase resumes with a fresh direction $d = -g$. Accordingly, iterations are partitioned into

$$\mathcal{N} = \{\text{descent steps for which (3.5) holds}\}, \qquad \mathcal{R} = \{\text{descent steps not in } \mathcal{N}\},$$
$$\mathcal{W} = \{\text{escape-window steps}\}.$$

where $\mathcal{W}$ collects the inner iterations of all windows, so that the total iteration count is $|\mathcal{N}| + |\mathcal{R}| + |\mathcal{W}|$. This partition drives the complexity analysis: $\mathcal{N}$- and $\mathcal{R}$-steps each yield a per-step decrease of f, with distinct constants $c_{\mathcal{N}}$ and $c_{\mathcal{R}}$ (Section 3.2), while $\mathcal{W}$-steps are accounted for through the escape guarantee (Sections 3.3 and 3.4). The three terms of Theorem 1 correspond exactly to these three sets.

3.1.4 Main result

Theorem 1 (Main result)**.** *Let Assumptions 1 to 3 hold, let $\varepsilon_g \in (0,1]$, $\varepsilon_H \in (0, L_1]$ and $\delta_{\mathrm{tot}} \in (0,1)$, and let the restart parameters satisfy $q \in [0, 1+p]$. Run Algorithm 1 with the parameters of Proposition 1 and the failure budget (3.57), and let ϕ_0 be the logarithmic factor defined in Section 3.4.3. Then Algorithm 1 terminates and returns a point $\tilde{x}$ which, with probability at least $1-\delta_{\mathrm{tot}}$, is an $(\varepsilon_g, \varepsilon_H)$-SOSP. The total number of iterations satisfies*

$$N \;\le\; \underbrace{\frac{\Delta_f}{c_{\mathcal{R}}}\,\varepsilon_g^{-2} + \frac{\Delta_f}{c_{\mathcal{N}}}\,\varepsilon_g^{-\max\{1+p,\,2(1+p-q)\}}}_{\text{descent phase (Section 3.2)}} + \underbrace{C\,\frac{L_1 L_2^2\,\Delta_f\,\phi_0^4}{\varepsilon_H^4}}_{\text{escape phase (Sections 3.3 and 3.4)}},$$

where $c_{\mathcal{N}}, c_{\mathcal{R}}$ are the line-search constants of Lemmas 1 and 2 and C depends only on the momentum γ and the analysis parameters c, ω. Since ϕ is logarithmic in the problem data, the escape term is $\widetilde{O}\big(L_1 L_2^2 \Delta_f\, \varepsilon_H^{-4}\big)$.

Remark 1 (Rates). Under the coupling $\varepsilon_H = \sqrt{L_2 \varepsilon_g}$ used in [27, 28], the escape term becomes $\widetilde{O}(L_1 \Delta_f\, \varepsilon_g^{-2})$, matching perturbed gradient descent [27] in both the ε_g- and the L_1-dependence.

Remark 2 (Iterations versus evaluations). Each escape-window iteration performs one gradient evaluation, plus one function evaluation per window for the certification test. A descent iteration performs one gradient evaluation and at most $\lfloor \bar{\jmath}_{\mathcal{N},k} + 1 \rfloor$ (resp. $\bar{\jmath}_{\mathcal{R}} + 1$) function evaluations for the backtracking line search (Lemmas 1 and 2).

Remark 3 (Modularity). The escape guarantee consumes nothing from the descent phase beyond the trigger $\|\nabla f(\tilde{x})\| \le \varepsilon_g$ and the reset $d = -g$: none of $(\sigma, \kappa, p, q, \eta, \theta)$ appears in Section 3.3 or in the window constants of Section 3.4. Any descent scheme providing a per-step decrease and the small-gradient trigger could occupy the descent phase.

3.2 The Descent Phase: Restarted NCG with Line Search

We recall the two decrease lemmas of [10] and derive a structural property of the safeguard used in Chapter 5. The lemmas apply verbatim here because our descent steps are exactly the restarted NCG steps of [10]: same restart test (3.5) and same backtracking line search, with the escape windows only interrupting the iteration and resetting $d = -g$ on re-entry (cf. Remark 3). They supply the per-step decrease and the line-search constants $c_{\mathcal{N}}, c_{\mathcal{R}}$ behind the descent-phase term

$$\frac{\Delta_f}{c_{\mathcal{R}}}\,\varepsilon_g^{-2} + \frac{\Delta_f}{c_{\mathcal{N}}}\,\varepsilon_g^{-\max\{1+p,\,2(1+p-q)\}}$$

of Theorem 1; the third lemma records the gradient floor used in Chapter 5.

Lemma 1 ([10, Lemma 3.1])**.** *Let Assumption 1 hold and $k \in \mathcal{N}$ with $\|g_k\| > 0$. The backtracking terminates after at most $\lfloor \bar{\jmath}_{\mathcal{N},k} + 1 \rfloor$ trials, with*

$$\bar{\jmath}_{\mathcal{N},k} := \big[\log_\theta\big(\tfrac{2(1-\eta)\sigma}{\kappa^2 L_1}\|g_k\|^{1+p-2q}\big)\big]_+,$$

and

$$f(x_k) - f(x_{k+1}) > c_{\mathcal{N}} \min\big\{\|g_k\|^{1+p},\, \|g_k\|^{2(1+p-q)}\big\}, \tag{3.7}$$

where

$$c_{\mathcal{N}} := \eta\sigma \min\Big\{1,\, \tfrac{2(1-\eta)\sigma\theta}{\kappa^2 L_1}\Big\}.$$

Lemma 2 ([10, Lemma 3.2])**.** *Let Assumption 1 hold and $k \in \mathcal{R}$ with $\|g_k\| > 0$. The backtracking terminates after at most $\bar{\jmath}_{\mathcal{R}} + 1$ trials, with*

$$\bar{\jmath}_{\mathcal{R}} := \big[\log_\theta\big(\tfrac{2(1-\eta)}{L_1}\big)\big]_+,$$

and

$$f(x_k) - f(x_{k+1}) > c_{\mathcal{R}}\|g_k\|^2, \tag{3.8}$$

where

$$c_{\mathcal{R}} := \eta \min\left\{1, \tfrac{2(1-\eta)\theta}{L_1}\right\}.$$

Both lemmas give $f(x_{k+1}) < f(x_k)$: f is strictly decreasing along descent steps. These per-step decreases drive the descent-phase count in Section 3.4.

Lemma 3 (Gradient threshold for non-restarted steps)**.** *Let $k \in \mathcal{N}$ with $\|g_k\| > 0$. Then the search direction is sandwiched,*

$$\sigma\,\|g_k\|^p \;\le\; \|d_k\| \;\le\; \kappa\,\|g_k\|^q, \tag{3.9}$$

and consequently the safeguard forces the scale inequality

$$\sigma\,\|g_k\|^{p-q} \;\le\; \kappa. \tag{3.10}$$

When $q > p$ the latter imposes a gradient floor*,*

$$\|g_k\| \;\ge\; (\sigma/\kappa)^{1/(q-p)}, \tag{3.11}$$

when $q < p$ it imposes a ceiling $\|g_k\| \le (\kappa/\sigma)^{1/(p-q)}$, and when $q = p$ it imposes no constraint on the gradient scale.

Proof. For $k \in \mathcal{N}$ the restart test (3.5) holds, so $g_k^\top d_k \le -\sigma\|g_k\|^{1+p}$ and hence $|g_k^\top d_k| \ge \sigma\|g_k\|^{1+p}$. Combining this with Cauchy–Schwarz and the norm bound in (3.5),

$$\sigma\|g_k\|^{1+p} \;\le\; |g_k^\top d_k| \;\le\; \|g_k\|\,\|d_k\| \;\le\; \kappa\,\|g_k\|^{1+q}.$$

Dividing the chain by $\|g_k\|^{1+p} > 0$ gives the lower bound in (3.9), whose upper bound is the second condition of (3.5); dividing instead by $\|g_k\|^{1+q} > 0$ gives the master inequality (3.10). Isolating $\|g_k\|$ according to the sign of $q - p$ then gives the floor (3.11) when $q > p$, the ceiling when $q < p$, and no constraint when $q = p$. □

Remark 4. Geometrically, (3.9) confines $\|d_k\|$ to the band $[\sigma\|g_k\|^p, \kappa\|g_k\|^q]$, which is empty below the floor (3.11) when $q > p$: there the method restarts by necessity and degrades to gradient descent. This is the mechanism behind Remark 11 in Chapter 5.

3.3 The Escape Phase: Perturbed Heavy-Ball Windows

This section is the core of this work. Following the perturbation strategy of [27, 28, 34], we show that after a perturbation near a strict saddle point, the HB window escapes the saddle with high probability. The argument shows that the set of "bad" perturbations, those from which the window fails to escape, is a thin slab of the perturbation ball, so that a uniformly random perturbation misses it with high probability.

The proof runs in three steps, spread across the subsections as follows.

Step 1 (geometric separation; Sections 3.3.2 to 3.3.4). We run two copies of the window from perturbations that differ only by a small displacement ℓ_0 along the escape direction $v_{\min}$. We show that the gap between the two trajectories grows geometrically in the number of window steps, at a rate $\mu_+ > 1$ set by the most negative curvature of the Hessian at the snapshot.

Step 2 (incompatibility with localization; Section 3.3.1 and Section 3.3.4). The HB energy confines a non-escaping trajectory to a ball of radius S around its start (Lemma 5). Two trajectories that both remain in such a ball can be at most $2S+\ell_0$ apart, which contradicts the geometric growth of Step 1 once the window is long enough. Hence at most one of the two trajectories can fail to escape.

Step 3 (volume argument; Section 3.3.5). Step 2 bounds the width of the "stuck" set along the escape direction. A volume estimate then shows that a uniformly random perturbation lands in this thin slab with low probability, which yields the escape guarantee.

3.3.1 Energy decrease and localization

Throughout this subsection we work inside a single escape window and use the associated notations: $\hat{x}_0, \dots, \hat{x}_T$ are the iterates of (3.6), $s_t := \hat{x}_{t+1} - \hat{x}_t$, and the start-from-rest convention $\hat{x}_{-1} := \hat{x}_0$ gives $s_{-1} = 0$. Inside the window the safeguard is off and f need not decrease; the monotone quantity we track is the classical HB energy [60, 41]

$$E_t \;:=\; f(\hat{x}_t) + \frac{\gamma}{2\alpha_{\mathrm{w}}} \|s_{t-1}\|^2. \tag{3.12}$$

Lemma 4 (Energy decrease along window steps). *Let Assumptions 1 and 3 hold, $\gamma \in (0, \frac{1}{2})$, and $\alpha_{\mathrm{w}} = 1/L_1$. Then for every window step $t \in \{0, \dots, T-1\}$, with $s_t = -\alpha_{\mathrm{w}} g_t + \gamma s_{t-1}$,*

$$E_{t+1} \;\le\; E_t - L_1\Big(\tfrac{1}{2} - \gamma\Big)\|s_t\|^2. \tag{3.13}$$

Moreover $E_t \ge f(\hat{x}_t) \ge f_{\mathrm{low}}$, and $s_{-1} = 0$ gives $E_0 = f(\hat{x}_0)$.

Proof. By L_1-smoothness,

$$f(\hat{x}_{t+1}) \le f(\hat{x}_t) + \langle g_t,\, s_t\rangle + \tfrac{L_1}{2}\|s_t\|^2.$$

The step identity $s_t = -\alpha_{\mathrm{w}} g_t + \gamma s_{t-1}$ rearranges to $g_t = \frac{1}{\alpha_{\mathrm{w}}}(\gamma s_{t-1} - s_t)$, so the linear term is

$$\langle g_t,\, s_t\rangle = \tfrac{\gamma}{\alpha_{\mathrm{w}}}\langle s_{t-1},\, s_t\rangle - \tfrac{1}{\alpha_{\mathrm{w}}}\|s_t\|^2.$$

Bounding $\langle s_{t-1},\, s_t\rangle \le \frac{1}{2}\|s_{t-1}\|^2 + \frac{1}{2}\|s_t\|^2$ by Young's inequality yields

$$f(\hat{x}_{t+1}) \le f(\hat{x}_t) + \frac{\gamma}{2\alpha_{\mathrm{w}}}\|s_{t-1}\|^2 - \Big(\frac{1}{\alpha_{\mathrm{w}}} - \frac{\gamma}{2\alpha_{\mathrm{w}}} - \frac{L_1}{2}\Big)\|s_t\|^2.$$

Adding $\frac{\gamma}{2\alpha_{\mathrm{w}}}\|s_t\|^2$ to both sides and using $\alpha_{\mathrm{w}} = 1/L_1$ gives (3.13). The bound $E_t \ge f(\hat{x}_t) \ge f_{\mathrm{low}}$ is immediate from (3.12) and Assumption 3, and $s_{-1} = 0$ gives $E_0 = f(\hat{x}_0)$. □

Remark 5 (Provenance and sharpness). E_t is the member $\nu = \gamma/2\alpha_{\mathrm{w}}$ of the Lyapunov family $H_\nu(x, y) = f(x) + \nu\|x - y\|^2$ used for the nonconvex HB method in [41, 60]. The threshold $\gamma < \frac{1}{2}$ is sharp for this family. Indeed, repeating the proof with a general coefficient $E_t^{(c)} = f(\hat{x}_t) + \frac{c}{2\alpha_{\mathrm{w}}}\|s_{t-1}\|^2$ and the weighted Young inequality $\langle s_{t-1},\, s_t\rangle \le \frac{1}{2\tau}\|s_{t-1}\|^2 + \frac{\tau}{2}\|s_t\|^2$, a telescoping decrease requires $c \ge \gamma/\tau$ and $c < 1 - \gamma\tau$; these are compatible for some $\tau > 0$ if and only if $\gamma < \frac{1}{2}$. The optimal choice $\tau = 1$, $c = \gamma$ recovers (3.12).

The energy bounds the total displacement over a window, which is the localization used against the geometric separation of Section 3.3.4.

Lemma 5 (Improve-or-localize). *Under the setting of Lemma 4, for every $0 \le t \le T$,*

$$\|\hat{x}_t - \hat{x}_0\| \;\le\; \sqrt{\frac{T\big(f(\hat{x}_0) - f(\hat{x}_T)\big)}{L_1(\frac{1}{2} - \gamma)}}. \tag{3.14}$$

In particular $f(\hat{x}_0) - f(\hat{x}_T) \le F$ implies $\|\hat{x}_t - \hat{x}_0\| \le S$ for all $t \le T$, where

$$S \;:=\; \sqrt{\frac{T F}{L_1(\frac{1}{2} - \gamma)}}. \tag{3.15}$$

Proof. Telescoping (3.13) and using monotonicity of E ($E_t \geq E_T$) together with $E_0 = f(\hat{x}_0)$,

$$\begin{aligned} L_1\Big(\tfrac{1}{2} - \gamma\Big) \sum_{j=0}^{t-1} \|s_j\|^2 &\leq E_0 - E_t \\ &\leq E_0 - E_T \\ &= f(\hat{x}_0) - f(\hat{x}_T) - \frac{\gamma}{2\alpha_{\mathrm{w}}}\|s_{T-1}\|^2 \\ &\leq f(\hat{x}_0) - f(\hat{x}_T). \end{aligned}$$

By the triangle inequality and Cauchy–Schwarz, $\|\hat{x}_t - \hat{x}_0\| \leq \sqrt{t}\big(\sum_{j<t}\|s_j\|^2\big)^{1/2}$, and $t \leq T$ gives (3.14). $\square$

Lemma 5 is a dichotomy: if the window fails to decrease f by more than F, i.e. $f(\hat{x}_0) - f(\hat{x}_T) \leq F$, then every iterate stays within distance S of the start (*localize*). Equivalently, leaving the ball of radius S forces a function decrease exceeding F (*improve*). The escape analysis exploits the first statement: a non-escaping trajectory is confined to $B^{(n)}_{\hat{x}_0}(S)$.

3.3.2 The coupled difference recursion

We now set up the two-trajectory scheme announced in Step 1. Fix a snapshot $\tilde{x}$ at which a perturbation is added, write $H := \nabla^2 f(\tilde{x})$, and let $v_{\min}$ be a unit eigenvector associated with $\lambda_{\min}(H)$ (the escape direction). Consider two runs of the same window, started from seeds that differ only along $v_{\min}$:

$$\hat{x}_0 = \tilde{x} + \xi, \qquad \hat{x}'_0 = \tilde{x} + \xi', \qquad \xi' = \xi + \ell_0\, v_{\min}, \qquad \xi, \xi' \in B_0(R),$$

and set $\Delta x_t := \hat{x}_t - \hat{x}'_t$ and $\Delta d_t := d_t - d'_t$. Both runs obey the start-from-rest convention $d_{-1} = d'_{-1} = 0$ of Section 3.3.1, so the reset $d_0 = -g_0$ is the $t = 0$ step of each window. The rescaled difference state is

$$w_t := \begin{pmatrix} \Delta x_t \\ \alpha_{\mathrm{w}}\, \Delta d_{t-1} \end{pmatrix} \in \mathbb{R}^{2n}, \qquad w_0 = \begin{pmatrix} -\ell_0\, v_{\min} \\ 0 \end{pmatrix}, \qquad \|w_0\| = \ell_0. \tag{3.16}$$

Note that w_0 is aligned with the escape direction: its first component is a multiple of $v_{\min}$ and its second vanishes. This alignment places w_0 in the two-dimensional invariant plane of the block associated with $\lambda_{\min}$, and is used in Lemma 11. Finally, write the gradient as its linear part at the snapshot plus a remainder, $g_t = H\hat{x}_t + e_t$ with $e_t := \nabla f(\hat{x}_t) - H\hat{x}_t$.

Lemma 6 (Difference recursion)**.** *Along an escape window, with constant stepsize $\alpha_{\mathrm{w}} = 1/L_1$ and constant momentum γ, the difference state obeys*

$$w_{t+1} = \mathbf{M}\, w_t + u_t, \qquad \mathbf{M} = \begin{pmatrix} I - \alpha_{\mathrm{w}} H & \gamma I \\ -\alpha_{\mathrm{w}} H & \gamma I \end{pmatrix}, \qquad u_t = \begin{pmatrix} -\alpha_{\mathrm{w}}\, \Delta e_t \\ -\alpha_{\mathrm{w}}\, \Delta e_t \end{pmatrix}, \tag{3.17}$$

where $\Delta e_t = \nabla f(\hat{x}_t) - \nabla f(\hat{x}'_t) - H\Delta x_t$. If moreover both runs remain localized in the sense of Lemma 5, that is

$$\|\hat{x}_t - \hat{x}_0\| \leq S \quad \textit{and} \quad \|\hat{x}'_t - \hat{x}'_0\| \leq S \qquad \textit{for all } 0 \leq t \leq T,$$

then the remainder is proportional to the gap:

$$\|u_t\| \leq \chi\, \|w_t\|, \qquad \chi := \frac{\sqrt{2}\, L_2\, (S + R)}{L_1}. \tag{3.18}$$

Proof. From $\hat{x}_{t+1} = \hat{x}_t + \alpha_{\mathrm{w}} d_t$ and $d_t = -(H\hat{x}_t + e_t) + \gamma d_{t-1}$, differencing the two runs gives

$$\alpha_{\mathrm{w}} \Delta d_t = -\alpha_{\mathrm{w}} H \Delta x_t - \alpha_{\mathrm{w}} \Delta e_t + \gamma\,(\alpha_{\mathrm{w}} \Delta d_{t-1}),$$
$$\Delta x_{t+1} = \Delta x_t + \alpha_{\mathrm{w}} \Delta d_t = (I - \alpha_{\mathrm{w}} H)\Delta x_t + \gamma\,(\alpha_{\mathrm{w}} \Delta d_{t-1}) - \alpha_{\mathrm{w}} \Delta e_t,$$

which is exactly (3.17). The stepsize α_{w} and the momentum γ are the *same* in both runs, so they factor out of the differences; this is what makes $\mathbf{M}$ constant and independent of t.

For the remainder bound, apply the fundamental theorem of calculus to ∇f along the segment joining $\hat{x}'_t$ to $\hat{x}_t$:

$$\Delta e_t = \Big(\int_0^1 \big(\nabla^2 f(\hat{x}'_t + r\,\Delta x_t) - H \big)\,\mathrm{d}r \Big) \Delta x_t,$$

and by Assumption 2, $\|\nabla^2 f(\hat{x}'_t + r\Delta x_t) - H\| \le L_2 \|\hat{x}'_t + r\Delta x_t - \tilde{x}\|$ for every $r \in [0,1]$. The map $r \mapsto \|\hat{x}'_t + r\Delta x_t - \tilde{x}\|$ is convex, so it attains its maximum over $[0,1]$ at an endpoint, that is at $\hat{x}_t$ or $\hat{x}'_t$. Both endpoints are bounded by the localization hypothesis and $\xi \in B_0(R)$:

$$\|\hat{x}_t - \tilde{x}\| \le \|\hat{x}_t - \hat{x}_0\| + \|\hat{x}_0 - \tilde{x}\| \le S + R,$$

and identically for the primed run. Hence

$$\|\Delta e_t\| \le \Big(\int_0^1 L_2\, \|\hat{x}'_t + r\Delta x_t - \tilde{x}\|\,\mathrm{d}r \Big) \|\Delta x_t\| \le L_2\,(S+R)\,\|\Delta x_t\|.$$

Finally, both components of u_t equal $-\alpha_{\mathrm{w}} \Delta e_t$, so $\|u_t\| = \sqrt{2}\,\alpha_{\mathrm{w}} \|\Delta e_t\|$, and Δx_t is the first block of w_t, so $\|\Delta x_t\| \le \|w_t\|$. Combining with $\alpha_{\mathrm{w}} = 1/L_1$ gives (3.18). □

Remark 6 (Why the escape window uses constant α_{w} and γ). The form of (3.17) dictates two design choices in Algorithm 1: inside an escape window the stepsize and the momentum coefficient are constant, and the restart test (3.5) is suspended. Both are forced by the coupling for two reasons.

The remainder must be proportional to the gap. Suppose the two runs used per-iteration stepsizes α_t and α'_t. Differencing $\hat{x}_{t+1} = \hat{x}_t + \alpha_t d_t$ and $\hat{x}'_{t+1} = \hat{x}'_t + \alpha'_t d'_t$ gives

$$\Delta x_{t+1} = \Delta x_t + \alpha_t\,\Delta d_t + (\alpha_t - \alpha'_t)\,d'_t,$$

and the last term involves the *full* search direction d'_t. Its magnitude is of order $\|d'_t\|$, whereas every other term in the recursion is of order $\|w_t\|$, that is, of order of the gap between the trajectories. Consequently the bound $\|u_t\| \le \chi\|w_t\|$ of (3.18) fails, and with it the induction of Lemma 12, at its first step. A per-run momentum coefficient produces the same obstruction with the term $(\gamma_t - \gamma'_t)\,d'_{t-1}$, and the restart test is the extreme case: when the two runs make opposite restart decisions the coefficient jumps between β_t and 0, so $\beta_t - \beta'_t$ is as large as β_t itself and bears no relation to the gap. Suspending the test for the T steps following a perturbation makes the no-restart property an algorithmic fact for *both* runs, including trajectories that fail to escape.

$\mathbf{M}$ *must be constant.* Even with parameters shared between the two runs, letting them vary in t would replace $\mathbf{M}$ by a sequence $\mathbf{M}_t$, and the homogeneous part of the recursion by the product $\mathbf{M}_{T-1} \cdots \mathbf{M}_0$. The analysis of Section 3.3.3 bounds powers of a *fixed* matrix through its eigenstructure (Lemmas 7 and 10); products of distinct matrices admit no such description, and neither the uniform rate μ_+ nor the two-sided estimate of Lemma 11 survives.

Remark 7 (Open questions). Both obstructions above are limitations of the present analysis. Controlling a variable stepsize would require bounding $|\alpha_t - \alpha'_t|$ in terms of the gap $\|w_t\|$ itself, which is plausible for a line search that is Lipschitz in its starting point, but not available from the Armijo rule as stated. Controlling a variable momentum coefficient raises the same question for $|\beta_t - \beta'_t|$, which is what led us to abandon an initial design running restarted NCG in both phases. Accommodating a restart inside the window would instead require proving that the two runs restart simultaneously, or that neither restarts during the window. We leave these questions open.

3.3.3 Spectral analysis of the matrix $\mathbf{M}$

The recursion (3.17) is a linear system driven by the matrix $\mathbf{M}$ and perturbed by the remainder u_t. We need two bounds on the linear part. First, the homogeneous solution started from w_0 must grow at a geometric rate. Second, no solution may grow faster than that same rate. Both bounds must carry explicit constants, and both come from the eigenstructure of $\mathbf{M}$.

For gradient descent the matrix is $I - \alpha_{\mathrm{w}} H$, which is symmetric. It diagonalizes in the eigenbasis of H, and its eigenvalues give both bounds directly. Here the difference state carries a direction as well as a position, and $\mathbf{M}$ is not symmetric, so it does not diagonalize in that basis. However, along the eigenbasis of H it splits into 2×2 blocks, one per eigenvalue. The rest of the analysis works with those blocks.

Lemma 7 (Block reduction)**.** *Let $H = VDV^\top$ with V orthogonal and $D = \mathrm{diag}(\lambda_1, \dots, \lambda_n)$. In the reordered basis grouping the i-th coordinates of the two components of (3.16), $\mathbf{M}$ is orthogonally similar to* $\mathrm{blockdiag}(B_1, \dots, B_n)$ *with*

$$B_i = \begin{pmatrix} 1 - \alpha_{\mathrm{w}}\lambda_i & \gamma \\ -\alpha_{\mathrm{w}}\lambda_i & \gamma \end{pmatrix}, \qquad \det B_i = \gamma, \qquad \mathrm{tr}\, B_i = 1 - \alpha_{\mathrm{w}}\lambda_i + \gamma. \tag{3.19}$$

In particular, since the similarity is orthogonal,

$$\|\mathbf{M}^m\| = \max_i \|B_i^m\| \qquad \textit{for every } m \geq 0. \tag{3.20}$$

Proof. Set $P := \mathrm{diag}(V, V) \in \mathbb{R}^{2n \times 2n}$, which is orthogonal. Recall from (3.16) that $w = (\Delta x,\ \alpha_{\mathrm{w}} \Delta d)$, so the first block of $2n$ coordinates carries the position difference and the second carries the rescaled direction difference. Conjugating $\mathbf{M}$ by P applies $V^\top(\cdot)V$ to each $n \times n$ sub-block, sending H to D and I to I; every sub-block thereby becomes diagonal:

$$P^\top \mathbf{M}\, P = \begin{pmatrix} V^\top(I - \alpha_{\mathrm{w}} H)V & \gamma\, V^\top V \\ -\alpha_{\mathrm{w}}\, V^\top H V & \gamma\, V^\top V \end{pmatrix} = \begin{pmatrix} I - \alpha_{\mathrm{w}} D & \gamma I \\ -\alpha_{\mathrm{w}} D & \gamma I \end{pmatrix}. \tag{3.21}$$

In the new coordinates, write $\Delta x^{(i)}$ and $\alpha_{\mathrm{w}} \Delta d^{(i)}$ for the components of $V^\top \Delta x$ and $\alpha_{\mathrm{w}} V^\top \Delta d$ along the eigenvector v_i of H. Every sub-block of (3.21) being diagonal, the pair $(\Delta x^{(i)},\ \alpha_{\mathrm{w}} \Delta d^{(i)})$ evolves independently of all other indices: the only nonzero entries of $P^\top \mathbf{M} P$ couple these two coordinates to each other. Let Π be the permutation matrix reordering the $2n$ coordinates into the consecutive pairs $(i,\ n+i)$, $i = 1, \dots, n$. Then $P\Pi$ is orthogonal and gathers the coupled entries into n diagonal 2×2 blocks:

$$(P\Pi)^\top \mathbf{M}\, (P\Pi) = \mathrm{blockdiag}(B_1, \dots, B_n), \qquad B_i = \begin{pmatrix} 1 - \alpha_{\mathrm{w}}\lambda_i & \gamma \\ -\alpha_{\mathrm{w}}\lambda_i & \gamma \end{pmatrix}. \tag{3.22}$$

A direct computation gives $\det B_i = \gamma(1 - \alpha_{\mathrm{w}}\lambda_i) + \gamma\, \alpha_{\mathrm{w}}\lambda_i = \gamma$ and $\mathrm{tr}\, B_i = 1 - \alpha_{\mathrm{w}}\lambda_i + \gamma$, which is (3.19). Since the spectral norm is invariant under orthogonal similarity and the block-diagonal form has $\mathrm{blockdiag}(B_1, \dots, B_n)^m = \mathrm{blockdiag}(B_1^m, \dots, B_n^m)$, we obtain (3.20). □

Throughout the spectral analysis we treat the blocks of Lemma 7 as a one-parameter family. Since $\|H\| \leq L_1$ by Assumption 1, every eigenvalue of H lies in $[\lambda_{\min}, L_1]$, and for λ in that range we set

$$B(\lambda) = \begin{pmatrix} 1 - \alpha_{\mathrm{w}}\lambda & \gamma \\ -\alpha_{\mathrm{w}}\lambda & \gamma \end{pmatrix},$$

so that, by Lemma 7, $B(\lambda)$ has constant determinant and λ-dependent trace

$$\det B(\lambda) = \gamma, \qquad t(\lambda) := \mathrm{tr}\, B(\lambda) = 1 - \alpha_{\mathrm{w}}\lambda + \gamma,$$

characteristic polynomial $\mu^2 - t(\lambda)\mu + \gamma$, and discriminant

$$\Delta(\lambda) \coloneqq t(\lambda)^2 - 4\gamma.$$

When $\Delta(\lambda) \geq 0$ the eigenvalues are real, $\mu_\pm(\lambda) = \frac{1}{2}\big(t(\lambda) \pm \sqrt{\Delta(\lambda)}\big)$; when $\Delta(\lambda) < 0$ they are complex conjugates of modulus $\sqrt{\det B(\lambda)} = \sqrt{\gamma}$. We write $\rho(\lambda)$ for the spectral radius of $B(\lambda)$, so that $|\mu_+(\lambda) - \mu_-(\lambda)| = \sqrt{|\Delta(\lambda)|}$ in the real case. The block driving the escape is $B_{\min} \coloneqq B(\lambda_{\min})$, whose eigenvalues we abbreviate $\mu_+ \coloneqq \mu_+(\lambda_{\min})$ and $\mu_- \coloneqq \mu_-(\lambda_{\min})$; in particular $\mu_+\mu_- = \gamma$ and $t(\lambda_{\min}) = 1 + \gamma + \alpha_{\mathrm{w}}|\lambda_{\min}|$.

One elementary fact is used repeatedly: t is strictly decreasing in λ (since $\alpha_{\mathrm{w}} > 0$) and remains positive on the whole range, because with $\alpha_{\mathrm{w}} = 1/L_1$,

$$t(\lambda) \geq t(L_1) = 1 - \alpha_{\mathrm{w}} L_1 + \gamma = \gamma > 0 \qquad (\lambda \in [\lambda_{\min}, L_1]). \tag{3.23}$$

In the real regime both roots are therefore positive and $\rho(\lambda) = \mu_+(\lambda)$.

Bounding powers of these blocks at a prescribed geometric rate is a purely linear-algebraic matter, deferred to Chapter A. We use two bounds there, according to whether the block's eigenvalues are separated or nearly confluent. The first, Lemma 24, applies to any 2×2 matrix B with distinct eigenvalues $\mu_1 \neq \mu_2$ and spectral radius ρ, and reads

$$\|B^m\| \;\leq\; C(B)\,\rho^m, \qquad C(B) \coloneqq \frac{2\big(\|B\| + \rho\big)}{|\mu_1 - \mu_2|}, \tag{3.24}$$

so its constant blows up as the eigenvalue gap closes. The second, Lemma 25, bounds $\|B^m\|$ through a Schur form and never divides by the gap, at the price of a factor m; it is stated in (A.3).

The block associated with the most negative curvature drives the escape, and must have a real eigenvalue strictly larger than one for geometric growth to be possible.

Lemma 8 (Escape eigenvalue)**.** *Let $\lambda_{\min} \leq -\varepsilon_H$ and let $p(\mu) = \mu^2 - t\mu + \gamma$ be the characteristic polynomial of $B_{\min}$, with $t = 1 + \gamma + \alpha_{\mathrm{w}}|\lambda_{\min}|$. Then $B_{\min}$ has real eigenvalues $\mu_- < 1 < \mu_+$, and*

$$\mu_+ \;\geq\; 1 + \alpha_{\mathrm{w}}|\lambda_{\min}| \;\geq\; 1 + \alpha_{\mathrm{w}}\varepsilon_H. \tag{3.25}$$

Proof. Write $a \coloneqq \alpha_{\mathrm{w}}|\lambda_{\min}| > 0$, so that $t = 1 + \gamma + a$. Evaluating at 1,

$$p(1) = 1 - t + \gamma = -a < 0.$$

Since p is a monic quadratic, a strictly negative value at 1 places 1 strictly between its two roots; in particular both roots are real and $\mu_- < 1 < \mu_+$. Evaluating at $1 + a$,

$$p(1+a) = (1+a)\big[(1+a) - t\big] + \gamma = (1+a)(-\gamma) + \gamma = -\gamma a \leq 0,$$

so $1 + a$ also lies between the roots, and $\mu_+ \geq 1 + a = 1 + \alpha_{\mathrm{w}}|\lambda_{\min}|$. The second inequality in (3.25) follows from $|\lambda_{\min}| \geq \varepsilon_H$. $\square$

By Lemma 7 the spectral radius of $\mathbf{M}$ is $\max_i \rho(B_i)$. The next lemma shows this maximum is attained at the escape block, so that $\rho(\mathbf{M}) = \mu_+$.

Lemma 9 (No block grows faster)**.** *For every $\lambda \in [\lambda_{\min}, L_1]$, the spectral radius of $B(\lambda)$ satisfies $\rho(\lambda) \leq \mu_+$.*

Proof. If $\Delta(\lambda) < 0$, the eigenvalues are complex conjugates of modulus $\rho(\lambda) = \sqrt{\gamma} < 1 < \mu_+$, using $\gamma < \frac{1}{2}$ and Lemma 8.

If $\Delta(\lambda) \geq 0$, the eigenvalues are real and, by (3.23), positive; moreover $t(\lambda)^2 \geq 4\gamma$ and $t(\lambda) > 0$ give $t(\lambda) \geq 2\sqrt{\gamma}$, so

$$\rho(\lambda) = \mu_+(\lambda) = \tfrac{1}{2}\big(t(\lambda) + \sqrt{t(\lambda)^2 - 4\gamma}\big).$$

On $\{t \geq 2\sqrt{\gamma}\}$ the map $t \mapsto \frac{1}{2}\big(t + \sqrt{t^2 - 4\gamma}\big)$ is increasing, being a sum of increasing functions of t. Since t is strictly decreasing in λ, it attains its maximum over $[\lambda_{\min}, L_1]$ at $\lambda_{\min}$, where $t(\lambda_{\min}) \geq 1 + \gamma \geq 2\sqrt{\gamma}$ by the AM–GM inequality. Hence $\mu_+(\lambda) \leq \mu_+(\lambda_{\min}) = \mu_+$. □

Combining Lemmas 9 and 24 would give $\|\mathbf{M}^m\| \leq \max_i C(B_i)\, \mu_+^m$, but this is useless: the constant $C(B)$ of (3.24) divides by the gap $|\mu_1 - \mu_2|$ between the two eigenvalues, and that gap can be zero. It vanishes when $\Delta(\lambda) = 0$, that is at $\lambda^* = L_1(1 - \sqrt{\gamma})^2$, where both eigenvalues equal $\sqrt{\gamma}$. So $\max_i C(B_i)$ is unbounded over the Hessians we must handle.

At λ^* the obstruction is not that the bound of Lemma 24 is loose but that no bound at rate ρ exists at all. Since its off-diagonal entry γ is nonzero, $B(\lambda^*)$ is not scalar and is therefore defective, hence similar to a Jordan block, whose powers

$$\begin{pmatrix} \mu & 1 \\ 0 & \mu \end{pmatrix}^m = \begin{pmatrix} \mu^m & m\mu^{m-1} \\ 0 & \mu^m \end{pmatrix}$$

carry a factor $m\mu^{m-1}$ off the diagonal. Consequently $\|B(\lambda^*)^m\|$ is of order $m\rho^{m-1}$ with $\rho = \sqrt{\gamma}$, so $\|B(\lambda^*)^m\|/\rho^m \to \infty$ and no constant C can satisfy $\|B(\lambda^*)^m\| \leq C\rho^m$ for all m.

The situation is saved by the location of the degeneracy. It occurs only at $\lambda^* > 0$, where the spectral radius $\sqrt{\gamma}$ is bounded below one, whereas the target rate satisfies $\mu_+ > 1$. Bounding every block at the common rate μ_+ rather than at its own ρ absorbs the polynomial factor, since $m\rho^{m-1} \leq C(\gamma)\, \mu_+^m$ whenever $\rho < 1 < \mu_+$. Lemma 10 makes this precise, with a constant depending on γ and no division by the eigenvalue gap.

Lemma 10 (Uniform bound on powers of $\mathbf{M}$)**.** *Let $\gamma \in (0, 1)$ (in particular any $\gamma < \frac{1}{2}$ as used in Algorithm 1) and $\lambda_{\min} < 0$. Set*

$$\varsigma := \frac{1 - \gamma}{2}, \qquad \bar{\rho} := \frac{\sqrt{4\gamma + \varsigma^2} + \varsigma}{2}, \qquad b_{\mathbf{M}} := \max_i \|B_i\|_F.$$

Then $\bar{\rho} < 1$, $b_{\mathbf{M}} \leq \sqrt{5 + 2\gamma^2}$, and for every $m \geq 0$,

$$\|\mathbf{M}^m\| \leq C_{\mathbf{M}}\, \mu_+^m, \qquad C_{\mathbf{M}} := \max\left\{ \frac{4(b_{\mathbf{M}} + \mu_+)}{1 - \gamma},\ \sqrt{2} + \frac{b_{\mathbf{M}}}{(1 - \bar{\rho})^2} \right\}. \tag{3.26}$$

Proof. By (3.20) it suffices to bound $\|B(\lambda)^m\| \leq C_{\mathbf{M}}\mu_+^m$ for each eigenvalue $\lambda \in [\lambda_{\min}, L_1]$.

The bound of Lemma 24 is useless when the two eigenvalues are close, since its constant divides by $|\mu_1 - \mu_2| = |\Delta(\lambda)|^{1/2}$. We therefore split the spectrum at the threshold ς: blocks with gap at least ς are handled by Lemma 24, while blocks with smaller gap, which occur only at $\lambda > 0$ where $\rho < 1$, are handled through a Schur form that never divides by the gap.

A bound on $b_{\mathbf{M}}$ comes naturally from the parameters: since $\alpha_{\mathrm{w}} = 1/L_1$ and $|\lambda_i| \leq L_1$, we have $|\alpha_{\mathrm{w}}\lambda_i| \leq 1$, so

$$\|B_i\|_F^2 = (1 - \alpha_{\mathrm{w}}\lambda_i)^2 + (\alpha_{\mathrm{w}}\lambda_i)^2 + 2\gamma^2 \leq 4 + 1 + 2\gamma^2 = 5 + 2\gamma^2.$$

Likewise, $\bar{\rho} < 1$ is immediate. Indeed, as $2 - \varsigma = \frac{3+\gamma}{2} > 0$, both sides of the defining inequality may be squared:

$$\begin{aligned} \bar{\rho} < 1 &\iff \sqrt{4\gamma + \varsigma^2} < 2 - \varsigma \\ &\iff 4\gamma < 4 - 4\varsigma \end{aligned}$$

$$\begin{aligned}&\iff \gamma < 1 - \varsigma = \tfrac{1+\gamma}{2}\\&\iff \gamma < 1,\end{aligned}$$

which holds by hypothesis.

We now proceed to identify that the degeneracy sits at $\lambda > 0$. If $\lambda \le 0$ then $t(\lambda) \ge 1+\gamma$, so

$$\Delta(\lambda) \ge (1+\gamma)^2 - 4\gamma = (1-\gamma)^2 = 4\varsigma^2 \ge \varsigma^2. \tag{3.27}$$

Every block with $\lambda \le 0$, and in particular the escape block $B_{\min}$, therefore has gap at least ς, so the near-degenerate regime can occur only at $\lambda > 0$. The rest of the proof handles both cases.

Case 1 (large gap): $|\Delta(\lambda)| \ge \varsigma^2$. The eigenvalues are distinct, so Lemma 24 applies. With $\|B\| \le \|B\|_F \le b_{\mathbf{M}}$, the bound $\rho(\lambda) \le \mu_+$ of Lemma 9, and $|\mu_1 - \mu_2| = \sqrt{|\Delta(\lambda)|} \ge \varsigma$,

$$\begin{aligned}\|B^m\| &\le \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|}\,\rho^m\\&\le \frac{2(b_{\mathbf{M}} + \mu_+)}{\varsigma}\,\mu_+^m\\&= \frac{4(b_{\mathbf{M}} + \mu_+)}{1-\gamma}\,\mu_+^m.\end{aligned} \tag{3.28}$$

Case 2 (small gap): $|\Delta(\lambda)| < \varsigma^2$. By (3.27) this forces $\lambda > 0$. We first show that the spectral radius is bounded away from 1, then bound $\|B^m\|$ without dividing by the gap.

(a) The spectral radius satisfies $\rho \le \bar\rho < 1$. Two cases, according to the sign of $\Delta(\lambda)$.

If $\Delta(\lambda) \le 0$, the eigenvalues are complex conjugate of modulus $\rho = \sqrt{\gamma}$, and since $\varsigma^2 \ge 0$,

$$\bar\rho = \frac{\sqrt{4\gamma + \varsigma^2} + \varsigma}{2} \;\ge\; \frac{2\sqrt{\gamma}}{2} \;=\; \sqrt{\gamma} \;=\; \rho.$$

If $0 < \Delta(\lambda) < \varsigma^2$, the eigenvalues are real and positive by (3.23), so $t(\lambda) = \sqrt{\Delta(\lambda) + 4\gamma}$. Both terms of ρ are then bounded above:

$$t(\lambda) < \sqrt{\varsigma^2 + 4\gamma}, \qquad \sqrt{\Delta(\lambda)} < \varsigma,$$

hence

$$\rho = \frac{t(\lambda) + \sqrt{\Delta(\lambda)}}{2} \;<\; \frac{\sqrt{4\gamma + \varsigma^2} + \varsigma}{2} \;=\; \bar\rho.$$

In both cases $\rho \le \bar\rho$, and $\bar\rho < 1$ by Step 1.

(b) Bounding $\|B^m\|$. The gap may now be arbitrarily small, so we replace Lemma 24 by Lemma 25, which passes through a Schur form and never divides by the gap. With $\|B\|_F \le b_{\mathbf{M}}$, (A.3) gives

$$\|B^m\| \;\le\; \sqrt{2}\,\rho^m + b_{\mathbf{M}}\,m\,\rho^{m-1}.$$

It remains to absorb the factor m. Since $\rho \le \bar\rho < 1$, the term $m\rho^{m-1}$ is one summand of the convergent series $\sum_{l\ge1} l\,\bar\rho^{\,l-1} = (1-\bar\rho)^{-2}$, so

$$m\,\rho^{m-1} \;\le\; \frac{1}{(1-\bar\rho)^2},$$

while $\rho^m \le 1 \le \mu_+^m$ because $\mu_+ > 1$. Combining the last three displays,

$$\|B^m\| \le \Big(\sqrt{2} + \frac{b_{\mathbf{M}}}{(1-\bar\rho)^2}\Big)\mu_+^m. \tag{3.29}$$

Bounds (3.28) and (3.29) cover every block for all $m \ge 1$; for $m = 0$ each side is at least $\|I\| = 1 \le C_{\mathbf{M}}$. Taking the larger of the two constants gives $\|B_i^m\| \le C_{\mathbf{M}}\mu_+^m$ for every i, and (3.26) follows from (3.20). $\square$

It remains to show that the homogeneous part actually achieves the rate μ_+. This is where the alignment of w_0 noted in Section 3.3.2 matters: because the perturbation difference lies along $v_{\min}$, the homogeneous evolution $h_t = \mathbf{M}^t w_0$ never leaves the two-dimensional plane associated with $\lambda_{\min}$, and reduces there to iteration of the single block $B_{\min}$. Within that plane w_0 has a nonzero component along the unstable eigenvector of $B_{\min}$, so growth at rate μ_+ occurs from the first step, with an explicit constant.

Lemma 11 (Homogeneous growth)**.** *Let w_0 be as in (3.16) and set $h_t := \mathbf{M}^t w_0$. Then h_t remains in the two-dimensional invariant plane associated with $\lambda_{\min}$, where its coordinate pair $h_t^{\min} \in \mathbb{R}^2$ evolves as $h_t^{\min} = B_{\min}^t w_0^{\min}$ with $w_0^{\min} = (-\ell_0, 0)$, and for every $t \geq 0$,*

$$|\Delta x_t^{\text{hom}}| \;\geq\; c_{\text{esc}}\, \mu_+^t \ell_0 \qquad \text{and} \qquad \|h_t\| \;\geq\; c_{\text{esc}}\, \mu_+^t \ell_0, \qquad c_{\text{esc}} := \frac{\mu_+ - \gamma}{\mu_+ - \mu_-}, \tag{3.30}$$

where Δx_t^{hom} denotes the first coordinate of $h_t^{\min}$. Moreover $\|h_t\| \leq c_h\, \mu_+^t \ell_0$ with $c_h := C(B_{\min}) \leq 2(b_{\mathbf{M}} + \mu_+)/(1-\gamma)$, where $C(\cdot)$ is as in (3.24).

Proof. **Reduction to the escape block.** By Lemma 7, the orthogonal change of basis $P\Pi$ turns $\mathbf{M}$ into $\operatorname{blockdiag}(B_1, \ldots, B_n)$, grouping the $2n$ coordinates into n pairs, with block i acting on $(\Delta x^{(i)}, \alpha_{\text{w}} \Delta d^{(i)})$.

Consider the pairs of w_0 in this basis. Its second component is $\alpha_{\text{w}} \Delta d_{-1} = 0$, so the second entry of every pair vanishes. Its first component is $\Delta x_0 = -\ell_0\, v_{\min}$ by (3.16), which has a single nonzero coordinate in the eigenbasis of H, the one along $v_{\min}$. Hence every pair of w_0 is zero except the one associated with $\lambda_{\min}$, which equals

$$w_0^{\min} = (-\ell_0, 0).$$

Since the block-diagonal form leaves each pair invariant, $h_t = \mathbf{M}^t w_0$ remains in that plane for all t, and its surviving pair evolves as

$$h_t^{\min} = B_{\min}^t\, w_0^{\min}.$$

Because $P\Pi$ is orthogonal, norms are preserved, so we may work in the plane throughout.

Eigen-decomposition of $w_0^{\min}$. By Lemma 8 the eigenvalues $\mu_- < 1 < \mu_+$ of $B_{\min}$ are real and distinct, so $\{v_+, v_-\}$ below is a basis of the plane. For $\mu \in \{\mu_+, \mu_-\}$, the second row of $(B_{\min} - \mu I)v = 0$ reads

$$\alpha_{\text{w}} \lambda_{\min}\, v^{(1)} = (\gamma - \mu)\, v^{(2)},$$

so we may take the unnormalized eigenvectors

$$v_\mu = \begin{pmatrix} \gamma - \mu \\ \alpha_{\text{w}} \lambda_{\min} \end{pmatrix}, \qquad \mu \in \{\mu_+, \mu_-\}.$$

The two eigenvectors have the same second entry. Writing $w_0^{\min} = \pi_+ v_+ + \pi_- v_-$ and reading the two coordinates of $w_0^{\min} = (-\ell_0, 0)$ in turn:

the second coordinate gives $\alpha_{\text{w}} \lambda_{\min} (\pi_+ + \pi_-) = 0$, hence

$$\pi_- = -\pi_+;$$

the first coordinate then gives

$$-\ell_0 = \pi_+(\gamma - \mu_+) + \pi_-(\gamma - \mu_-) = \pi_+\big[(\gamma - \mu_+) - (\gamma - \mu_-)\big] = \pi_+\,(\mu_- - \mu_+),$$

so that

$$|\pi_+| = \frac{\ell_0}{\mu_+ - \mu_-}. \tag{3.31}$$

Lower bound. Write $\zeta := \mu_-/\mu_+ \in (0,1)$. Applying $B^t_{\min}$ to $w^{\min}_0 = \pi_+(v_+ - v_-)$,

$$h^{\min}_t = \pi_+\mu^t_+\big(v_+ - \zeta^t v_-\big),$$

whose first coordinate is

$$\Delta x^{\text{hom}}_t = \pi_+\,\mu^t_+\Big[(\gamma-\mu_+) - \zeta^t(\gamma-\mu_-)\Big].$$

We bound the bracket from below. Since $\mu_+\mu_- = \gamma$ and $\mu_+ > 1 > \gamma$, we have $\mu_- = \gamma/\mu_+ < \gamma$, so

$$\gamma - \mu_+ < 0 \qquad \text{and} \qquad \gamma - \mu_- > 0.$$

Both terms of the bracket are therefore negative, and dropping the second one can only decrease its magnitude:

$$\Big|(\gamma-\mu_+) - \zeta^t(\gamma-\mu_-)\Big| \;\geq\; |\gamma-\mu_+| \;=\; \mu_+ - \gamma \qquad \text{for every } t \geq 0.$$

Together with (3.31) this gives

$$|\Delta x^{\text{hom}}_t| \;\geq\; \frac{\mu_+ - \gamma}{\mu_+ - \mu_-}\,\mu^t_+\ell_0 \;=\; c_{\text{esc}}\,\mu^t_+\ell_0.$$

Finally Δx^{hom}_t is a coordinate of $h^{\min}_t$, so $\|h_t\| = \|h^{\min}_t\| \geq |\Delta x^{\text{hom}}_t|$, which completes (3.30).

Upper bound. We bound the constant $C(B_{\min})$ of (3.24). Since $\lambda_{\min} \leq 0$ we have $t(\lambda_{\min}) \geq 1+\gamma$, so the eigenvalue gap satisfies

$$\mu_+ - \mu_- = \sqrt{\Delta(\lambda_{\min})} \;\geq\; \sqrt{(1+\gamma)^2 - 4\gamma} \;=\; 1-\gamma,$$

while $\|B_{\min}\| \leq \|B_{\min}\|_F \leq b_{\mathbf{M}}$. Hence

$$C(B_{\min}) = \frac{2\big(\|B_{\min}\| + \mu_+\big)}{\mu_+ - \mu_-} \;\leq\; \frac{2(b_{\mathbf{M}} + \mu_+)}{1-\gamma}.$$

Lemma 24 with $\rho(B_{\min}) = \mu_+$ then gives

$$\|h_t\| \leq \|B^t_{\min}\|\,\|w^{\min}_0\| \leq C(B_{\min})\,\mu^t_+\ell_0 = c_h\,\mu^t_+\ell_0.$$

□

This subsection delivers the two-sided estimate that drives Section 3.3.4: the homogeneous part of the difference grows at least like $c_{\text{esc}}\,\mu^t_+\ell_0$ for every t (Lemma 11), while powers of $\mathbf{M}$ are bounded by $C_{\mathbf{M}}\,\mu^m_+$ uniformly over the spectrum of H (Lemma 10). The same rate appears on both sides, which is what the induction of Lemma 12 requires.

3.3.4 Geometric growth of the gap

Lemmas 7 to 11 show that the homogeneous part of the gap grows geometrically. It remains to check that the accumulated remainder does not cancel that growth. Split the difference state as $w_t = h_t + z_t$, where

$$h_t = \mathbf{M}^t w_0, \qquad z_t = \sum_{\tau=0}^{t-1} \mathbf{M}^{t-1-\tau} u_\tau,$$

so that h_t is the homogeneous solution of Lemma 11 and z_t collects the remainders of (3.17) propagated forward.

Lemma 12 (Geometric growth survives)**.** *Consider an escape window in which both trajectories remain localized, that is* $\|\hat{x}_t - \hat{x}_0\| \le S$ *and* $\|\hat{x}'_t - \hat{x}'_0\| \le S$ *for all* $t \le T$*, so that* (3.18) *applies. Let* $c \in \big(0, \min\{1, c_{\mathrm{esc}}/c_h\}\big)$ *and suppose*

$$\chi \;\le\; \frac{c\, c_{\mathrm{esc}}\, \mu_+}{C_{\mathbf{M}}\, c_h\, (1+c)\, (T+1)} \;=:\; \frac{\Lambda}{T+1}. \tag{3.32}$$

Then $\|z_t\| \le c\, \|h_t\|$ *for every* $t \le T$*, and in particular*

$$\|\Delta x_T\| \;\ge\; (c_{\mathrm{esc}} - c\, c_h)\, \mu_+^T \ell_0. \tag{3.33}$$

Proof. We prove $\|z_t\| \le c\|h_t\|$ for all $t \le T$ by induction on t.

Base case. $z_0 = 0$, so the claim holds at $t = 0$.

Induction step. Let $0 \le t \le T-1$ and assume $\|z_\tau\| \le c\|h_\tau\|$ for every $\tau \le t$. For such τ, the decomposition $w_\tau = h_\tau + z_\tau$, the induction hypothesis and the upper bound $\|h_\tau\| \le c_h \mu_+^\tau \ell_0$ of Lemma 11 give

$$\|w_\tau\| \le (1+c)\|h_\tau\| \le (1+c)\, c_h\, \mu_+^\tau \ell_0. \tag{3.34}$$

Using (3.26) to propagate each remainder, (3.18) to bound it, and then (3.34),

$$\begin{aligned}
\|z_{t+1}\| \;&\le\; \sum_{\tau=0}^{t} \|\mathbf{M}^{t-\tau} u_\tau\| \;\le\; \sum_{\tau=0}^{t} C_{\mathbf{M}}\, \mu_+^{t-\tau}\, \chi\, \|w_\tau\| \\
&\le\; C_{\mathbf{M}}\, \chi\, (1+c)\, c_h\, \ell_0 \sum_{\tau=0}^{t} \mu_+^{t-\tau} \mu_+^\tau \\
&=\; C_{\mathbf{M}}\, \chi\, (1+c)\, c_h\, (t+1)\, \mu_+^t\, \ell_0.
\end{aligned}$$

Since $t \le T-1$ we have $t+1 \le T+1$, so (3.32) applies and yields

$$\|z_{t+1}\| \;\le\; C_{\mathbf{M}}\, c_h\, (1+c)\, (T+1)\, \chi\, \mu_+^t\, \ell_0 \;\le\; c\, c_{\mathrm{esc}}\, \mu_+^{t+1}\, \ell_0 \;\le\; c\, \|h_{t+1}\|,$$

the last inequality by the lower bound of Lemma 11. This closes the induction.

Conclusion. Both Δx_T and $\Delta x_T^{\mathrm{hom}}$ are the first blocks of w_T and h_T respectively, so their difference is the first block of z_T and has norm at most $\|z_T\|$. The reverse triangle inequality, (3.30) and $\|z_T\| \le c\|h_T\| \le c\, c_h \mu_+^T \ell_0$ then give

$$\|\Delta x_T\| \;\ge\; |\Delta x_T^{\mathrm{hom}}| - \|z_T\| \;\ge\; (c_{\mathrm{esc}} - c\, c_h)\, \mu_+^T \ell_0,$$

which is (3.33). The constraint $c < c_{\mathrm{esc}}/c_h$ makes this bound positive. □

The two trajectories therefore separate at the geometric rate μ_+, even when neither of them escapes. Section 3.3.5 turns this into the escape guarantee.

3.3.5 Escape with high probability

We now carry out Step 3. Lemma 12 says that two perturbations separated along $v_{\min}$ drift apart at rate μ_+, provided both trajectories stay localized. Localization caps how far apart they can be: two runs confined to balls of radius S are at most $2S + \ell_0$ apart. For large T these two facts are incompatible, which bounds the separation ℓ_0 between any two localized seeds. That bound makes the localized set thin in the $v_{\min}$ direction, and a volume computation turns thinness into a probability.

Fix a snapshot $\tilde{x}$ with $\|\nabla f(\tilde{x})\| \le \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \le -\varepsilon_H$, and let $\hat{x}_0, \dots, \hat{x}_T$ denote the window run from the seed ξ, that is from $\hat{x}_0 = \tilde{x} + \xi$. Define

$$\mathcal{X}_{\mathrm{loc}} := \big\{\xi \in B_0(R) : \; \|\hat{x}_t - \hat{x}_0\| \le S \;\text{ for all } 0 \le t \le T\big\}, \tag{3.35}$$

$$\mathcal{X}_{\text{stuck}} \coloneqq \big\{\xi \in B_0(R) : \ f(\hat{x}_T) - f(\hat{x}_0) > -F\big\}. \tag{3.36}$$

By Lemma 5, a seed whose window fails to decrease f by more than F keeps every iterate within S of $\hat{x}_0$, so

$$\mathcal{X}_{\text{stuck}} \subseteq \mathcal{X}_{\text{loc}}. \tag{3.37}$$

It therefore suffices to bound the probability of landing in $\mathcal{X}_{\text{loc}}$. Finally, decompose a seed along the escape direction and its orthogonal complement,

$$\xi = \xi_1 v_{\min} + \xi_\perp, \qquad \xi_1 = \langle \xi, v_{\min} \rangle \in \mathbb{R}, \qquad \xi_\perp \in v_{\min}^\perp \cong \mathbb{R}^{n-1}. \tag{3.38}$$

Lemma 13 (Separation threshold)**.** *Let the hypotheses of Lemma 12 hold and assume $(c_{\text{esc}} - c\,c_h)\,\mu_+^T > 1$. If ξ and $\xi' = \xi + \ell_0\, v_{\min}$ both belong to $\mathcal{X}_{\text{loc}}$, then*

$$\ell_0 \ \le\ \ell^\star \ \coloneqq\ \frac{2S}{(c_{\text{esc}} - c\,c_h)\,\mu_+^T - 1}\,. \tag{3.39}$$

Proof. The two runs start at $\hat{x}_0 = \tilde{x} + \xi$ and $\hat{x}_0' = \tilde{x} + \xi'$, so $\|\hat{x}_0 - \hat{x}_0'\| = \ell_0$. Both seeds lie in $\mathcal{X}_{\text{loc}}$, so each run stays within S of its own start, and the triangle inequality gives

$$\|\Delta x_T\| \ \le\ \|\hat{x}_T - \hat{x}_0\| + \|\hat{x}_0 - \hat{x}_0'\| + \|\hat{x}_0' - \hat{x}_T'\| \ \le\ 2S + \ell_0. \tag{3.40}$$

Both runs being localized, Lemma 12 applies and gives (3.33). Combining it with (3.40),

$$(c_{\text{esc}} - c\,c_h)\,\mu_+^T\,\ell_0 \ \le\ 2S + \ell_0,$$

that is, $\big[(c_{\text{esc}} - c\,c_h)\mu_+^T - 1\big]\ell_0 \le 2S$. The bracket is positive by hypothesis, so dividing by it yields (3.39). □

Lemma 13 says that $\mathcal{X}_{\text{loc}}$ contains no two points more than $\ell^\star$ apart along $v_{\min}$: it is a slab of width at most $\ell^\star$ in that direction. The next lemma converts this into a volume bound.

Lemma 14 (The localized set is thin)**.** *Under the hypotheses of Lemma 13,*

$$\mathbb{P}\big(\xi \in \mathcal{X}_{\text{loc}}\big) \ \le\ \frac{\ell^\star}{R}\sqrt{\frac{n+1}{2\pi}}, \qquad \xi \sim \mathrm{Unif}\big(B_0(R)\big). \tag{3.41}$$

Proof. Fix $\xi_\perp \in v_{\min}^\perp$ and consider the corresponding slice of the localized set,

$$I(\xi_\perp) \ \coloneqq\ \big\{a \in \mathbb{R} : \ a\, v_{\min} + \xi_\perp \in \mathcal{X}_{\text{loc}}\big\}.$$

Let $a, a' \in I(\xi_\perp)$ with $a < a'$. The two seeds $\xi = a\, v_{\min} + \xi_\perp$ and $\xi' = a' v_{\min} + \xi_\perp$ satisfy $\xi' = \xi + (a' - a) v_{\min}$ and both lie in $\mathcal{X}_{\text{loc}}$, so Lemma 13 gives $a' - a \le \ell^\star$. Hence $I(\xi_\perp)$ has diameter at most $\ell^\star$, and its one-dimensional Lebesgue measure satisfies $|I(\xi_\perp)| \le \ell^\star$.

Integrating the indicator of $\mathcal{X}_{\text{loc}}$ in the coordinates (3.38) and using this bound on each slice,

$$\mathrm{Vol}\big(\mathcal{X}_{\text{loc}}\big) = \int_{B_0^{(n-1)}(R)} |I(\xi_\perp)|\, \mathrm{d}\xi_\perp \ \le\ \ell^\star\, \mathrm{Vol}\big(B_0^{(n-1)}(R)\big),$$

where $B_0^{(n-1)}(R)$ is the ball of radius R in $v_{\min}^\perp$.

Since ξ is uniform on $B_0(R)$,

$$\mathbb{P}\big(\xi \in \mathcal{X}_{\text{loc}}\big) = \frac{\mathrm{Vol}(\mathcal{X}_{\text{loc}})}{\mathrm{Vol}(B_0(R))} \ \le\ \ell^\star\, \frac{\mathrm{Vol}(B_0^{(n-1)}(R))}{\mathrm{Vol}(B_0(R))}.$$

Using $\mathrm{Vol}\big(B_0^{(m)}(R)\big) = \pi^{m/2}R^m/\Gamma(\frac{m}{2}+1)$,

$$\frac{\mathrm{Vol}(B_0^{(n-1)}(R))}{\mathrm{Vol}(B_0(R))} = \frac{1}{R\sqrt{\pi}} \cdot \frac{\Gamma\big(\frac{n}{2}+1\big)}{\Gamma\big(\frac{n+1}{2}\big)} \;\le\; \frac{1}{R\sqrt{\pi}}\sqrt{\frac{n+1}{2}} \;=\; \frac{1}{R}\sqrt{\frac{n+1}{2\pi}},$$

where the inequality uses the property of the Gamma function that $\Gamma(x+1)/\Gamma(x+\frac{1}{2}) \le \sqrt{x+\frac{1}{2}}$ for $x \ge 0$, applied at $x = n/2$. Combining the last two displays gives (3.41). □

Lemma 15 (Escape with high probability). *Let $\|\nabla f(\tilde{x})\| \le \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \le -\varepsilon_H$, let the hypotheses of Lemma 12 hold, and let $\delta \in (0,1)$ and $\omega \in (0,1)$. Suppose the window length satisfies*

$$T \;\ge\; \frac{1}{\log \mu_+}\,\log\left[\frac{1}{c_{\mathrm{esc}} - c\,c_h}\left(1 + \frac{2S}{\delta R}\sqrt{\frac{n+1}{2\pi}}\right)\right], \tag{3.42}$$

and the perturbation radius satisfies

$$\varepsilon_g R + \tfrac{L_1}{2}R^2 \;\le\; \omega F, \qquad \textit{for which} \qquad R \;\le\; \min\left\{\frac{\omega F}{2\varepsilon_g},\, \sqrt{\frac{\omega F}{L_1}}\right\} \;\; \textit{suffices.} \tag{3.43}$$

Then, with probability at least $1-\delta$ over $\xi \sim \mathrm{Unif}(B_0(R))$,

$$f(\hat{x}_T) - f(\tilde{x}) \;\le\; -(1-\omega)F \;=\; -F_{\mathrm{cert}}. \tag{3.44}$$

Proof. Exponentiating (3.42) gives

$$(c_{\mathrm{esc}} - c\,c_h)\,\mu_+^T \;\ge\; 1 + \frac{2S}{\delta R}\sqrt{\frac{n+1}{2\pi}} \;>\; 1, \tag{3.45}$$

which is the hypothesis required by Lemma 13, so that both Lemma 13 and Lemma 14 apply.

We first bound the probability of landing in the stuck set. Rearranging (3.45),

$$(c_{\mathrm{esc}} - c\,c_h)\,\mu_+^T - 1 \;\ge\; \frac{2S}{\delta R}\sqrt{\frac{n+1}{2\pi}},$$

so the separation threshold of Lemma 13 obeys

$$\ell^\star = \frac{2S}{(c_{\mathrm{esc}} - c\,c_h)\mu_+^T - 1} \;\le\; \delta R\sqrt{\frac{2\pi}{n+1}}.$$

Substituting this into (3.41) and using the inclusion (3.37),

$$\mathbb{P}\big(\xi \in \mathcal{X}_{\mathrm{stuck}}\big) \;\le\; \mathbb{P}\big(\xi \in \mathcal{X}_{\mathrm{loc}}\big) \;\le\; \frac{\ell^\star}{R}\sqrt{\frac{n+1}{2\pi}} \;\le\; \delta.$$

It remains to show that a seed outside the stuck set produces the announced decrease. Suppose then that $\xi \notin \mathcal{X}_{\mathrm{stuck}}$, which by the bound just established occurs with probability at least $1-\delta$. The window itself decreases f by more than F, since by (3.36)

$$f(\hat{x}_T) - f(\hat{x}_0) \;\le\; -F,$$

while the perturbation increases it by at most ωF: applying L_1-smoothness at $\tilde{x}$ and using $\|\xi\| \le R$ together with $\|\nabla f(\tilde{x})\| \le \varepsilon_g$,

$$f(\hat{x}_0) - f(\tilde{x}) \;\le\; \langle \nabla f(\tilde{x}), \xi\rangle + \tfrac{L_1}{2}\|\xi\|^2 \;\le\; \varepsilon_g R + \tfrac{L_1}{2}R^2 \;\le\; \omega F,$$

the last inequality being (3.43). Adding the two displays,

$$f(\hat{x}_T) - f(\tilde{x}) = \big(f(\hat{x}_T) - f(\hat{x}_0)\big) + \big(f(\hat{x}_0) - f(\tilde{x})\big) \;\le\; -F + \omega F = -(1-\omega)F,$$

which is (3.44).

Finally, the condition stated as sufficient for (3.43) is indeed so: $R \le \omega F/(2\varepsilon_g)$ gives $\varepsilon_g R \le \omega F/2$, and $R \le \sqrt{\omega F/L_1}$ gives $\frac{L_1}{2}R^2 \le \omega F/2$, so their sum is at most ωF. □

Lemma 15 completes the escape phase: a window started at a strict saddle decreases f by at least F_{cert} with probability $1-\delta$, which is exactly the test (3.1) performed by Algorithm 1. It remains to choose T, S, R and F so that (3.15), (3.32), (3.42) and (3.43) hold simultaneously, and to count the iterations; this is done in Section 3.4.

3.4 Iteration Complexity

This section proves Theorem 1. Sections 3.2 and 3.3 each imposed conditions on the derived constants T, S, R, F and on the per-window failure probability δ; we first collect these into a single system (Section 3.4.1), then show the system is solvable and compute the resulting scales (Section 3.4.2), and finally count iterations against the budget Δ_f (Section 3.4.3).

The iteration count is entirely deterministic. It uses only the per-step decreases of Section 3.2 and the outcome of the certification test (3.1), both of which the algorithm evaluates as it runs. Lemma 15 plays no part in it, and enters only to guarantee that a point returned by the test really is second-order stationary.

3.4.1 The constraint system

We collect the conditions imposed by Section 3.3. The derived constants must satisfy the following, where the reference after each line gives its origin.

$$\text{(C1)}\quad F = L_1\Big(\tfrac{1}{2}-\gamma\Big)\frac{S^2}{T}, \qquad \text{(3.15), solved for } F \tag{3.46}$$

$$\text{(C2)}\quad \chi = \frac{\sqrt{2}\,L_2(S+R)}{L_1} \;\le\; \frac{\Lambda}{T+1}, \qquad \Lambda := \frac{c\,c_{\mathrm{esc}}\,\mu_+}{C_{\mathbf{M}}\,c_h\,(1+c)}, \qquad \text{(3.32), (3.18)} \tag{3.47}$$

$$\text{(C3)}\quad \varepsilon_g R + \tfrac{L_1}{2}R^2 \;\le\; \omega F, \qquad \text{(3.43)} \tag{3.48}$$

$$\text{(C4)}\quad T \;\ge\; \frac{1}{\log\mu_+}\log\Big[\tfrac{1}{c_{\mathrm{esc}}-c\,c_h}\Big(1+\tfrac{2S}{\delta R}\sqrt{\tfrac{n+1}{2\pi}}\Big)\Big]. \qquad \text{(3.42)} \tag{3.49}$$

Here, $c \in (0,\min\{1, c_{\mathrm{esc}}/c_h\})$ and $\omega \in (0,1)$ are free parameters of the analysis, and $c_{\mathrm{esc}}, c_h, C_{\mathbf{M}}, \mu_+$ are the spectral constants of Lemmas 10 and 11, depending only on γ and ε_H.

3.4.2 Solving the system

We proceed in two stages. First we express S, F and R as functions of the window length T, in such a way that (3.46)–(3.48) hold for *every* $T \ge 1$ (Lemma 16). This leaves only (3.49), which becomes a single inequality in T; we then check that it holds for all T large enough (Proposition 1). Throughout we write

$$\kappa_\gamma := L_1\Big(\tfrac{1}{2}-\gamma\Big) > 0.$$

Lemma 16 (Constants at fixed window length)**.** *For $T \ge 1$ define*

$$\Psi(T) := \frac{\Lambda\,L_1}{\sqrt{2}\,L_2\,(T+1)}, \qquad S(T) := \frac{\Psi(T)}{2}, \qquad F(T) := \kappa_\gamma\,\frac{S(T)^2}{T}, \tag{3.50}$$

$$R(T) := \min\Big\{\frac{\omega F(T)}{2\varepsilon_g},\ \sqrt{\frac{\omega F(T)}{L_1}}\Big\}. \tag{3.51}$$

Then $S(T), F(T), R(T) > 0$, and (3.46), (3.47) and (3.48) hold for every $T \ge 1$.

Proof. Positivity is trivial, and (3.46) holds by definition.

(3.48): the two entries of the minimum in (3.51) control one term each. From $R \le \omega F/(2\varepsilon_g)$ we get $\varepsilon_g R \le \frac{1}{2}\omega F$, and from $R \le \sqrt{\omega F/L_1}$ we get $\frac{L_1}{2}R^2 \le \frac{1}{2}\omega F$. Adding gives $\varepsilon_g R + \frac{L_1}{2}R^2 \le \omega F$.

(3.47): it says $S + R \le \Psi$, and since $S = \Psi/2$ this amounts to $R \le S$. Substituting $F = \kappa_\gamma S^2/T$ and $\kappa_\gamma/L_1 = \frac{1}{2} - \gamma$ into the second entry of (3.51),

$$\sqrt{\frac{\omega F(T)}{L_1}} = S(T)\sqrt{\frac{\omega\,(\frac{1}{2}-\gamma)}{T}} \;\le\; S(T), \tag{3.52}$$

because $\omega < 1$, $\frac{1}{2}-\gamma < \frac{1}{2}$ and $T \ge 1$. As R is a minimum, $R(T) \le S(T)$, so $S+R \le 2S = \Psi$. $\square$

It remains to satisfy (3.49). Substituting (3.50) and (3.51) turns its right-hand side into a function of T alone,

$$G(T) := \frac{1}{\log \mu_+}\,\log\left[\frac{1}{c_{\mathrm{esc}} - c\,c_h}\left(1 + \frac{2S(T)}{\delta\,R(T)}\sqrt{\frac{n+1}{2\pi}}\right)\right], \tag{3.53}$$

so that (3.49) reads $T \ge G(T)$.

Proposition 1 (Solvability)**.** *Let $\gamma \in (0, \frac{1}{2})$, $c \in \big(0, \min\{1, c_{\mathrm{esc}}/c_h\}\big)$, $\omega \in (0,1)$, $\varepsilon_g \in (0,1]$, $\varepsilon_H \in (0, L_1]$ and $\delta \in (0,1)$. Then $T \ge G(T)$ holds for all T large enough. For any such T, the constants $S(T), F(T), R(T)$ of (3.50)–(3.51) satisfy the full system (3.46)–(3.49).*

Proof. By Lemma 16 only (3.49), i.e. $T \ge G(T)$, remains, so it suffices to show that G grows logarithmically.

The variable T enters G only through the ratio S/R. Inverting the minimum in (3.51) and substituting $F = \kappa_\gamma S^2/T$,

$$\frac{S(T)}{R(T)} = \max\left\{\frac{2\varepsilon_g\,T}{\omega\,\kappa_\gamma\,S(T)},\ \sqrt{\frac{T}{\omega(\frac{1}{2}-\gamma)}}\right\}. \tag{3.54}$$

By (3.50), $1/S(T)$ is a linear function of T, so the first entry is a quadratic polynomial in T and the second is of order $\sqrt{T}$. Hence $S(T)/R(T) \le P(T)$ for some quadratic polynomial P with positive coefficients depending only on $\gamma, c, \omega, L_1, L_2$ and ε_g. The bracket in (3.53) is therefore at most a quadratic polynomial in T, and

$$G(T) \;=\; O(\log T) \qquad \text{as } T \to \infty.$$

Since $T - G(T) \to +\infty$, there is $T_0 \ge 1$ with $T \ge G(T)$ for all $T \ge T_0$, which is (3.49). $\square$

Proposition 1 settles existence: Algorithm 1 can be run with parameters meeting every requirement of Section 3.3. It says nothing about their size, which is needed to compute a complexity bound. We therefore take T to be the smallest admissible window length and evaluate the resulting orders of magnitude.

Lemma 17 (Bounds on the escape rate)**.** *Let $\gamma \in (0, \frac{1}{2})$, $\alpha_{\mathrm{w}} = 1/L_1$, $\varepsilon_H \in (0, L_1]$, and let $\lambda_{\min} \le -\varepsilon_H$. Set $a := \alpha_{\mathrm{w}}|\lambda_{\min}|$. Then $a \in [\alpha_{\mathrm{w}}\varepsilon_H, 1]$ and*

$$1 + a \;\le\; \mu_+ \;<\; 1 + \frac{a}{1-\gamma} \;<\; 3, \qquad \frac{a}{2} \;\le\; \log\mu_+ \;<\; \frac{a}{1-\gamma}. \tag{3.55}$$

In particular $\log\mu_+ \ge \varepsilon_H/(2L_1)$.

Proof. By Assumption 1, $|\lambda_{\min}| \le \|H\| \le L_1$, so $a \le 1$; and $a \ge \alpha_{\mathrm{w}}\varepsilon_H$ by hypothesis.

Factoring $p(\mu) = (\mu-\mu_+)(\mu-\mu_-)$ and evaluating at 1 gives $(1-\mu_+)(1-\mu_-) = p(1) = 1-t+\gamma = -a$, that is

$$(\mu_+ - 1)(1-\mu_-) = a.$$

By Lemma 8, $\mu_+ > 1$, so $\mu_- = \gamma/\mu_+ \in (0,\gamma)$ and hence $1-\gamma < 1-\mu_- < 1$. Dividing,

$$a \;<\; \mu_+ - 1 \;<\; \frac{a}{1-\gamma},$$

which with $a \le 1$ and $\frac{1}{1-\gamma} < 2$ gives the first chain in (3.55).

For the logarithm, $\log\mu_+ \le \mu_+ - 1 < a/(1-\gamma)$. Conversely $\log\mu_+ \ge \log(1+a) \ge a/2$, since $h(u) = \log(1+u) - u/2$ satisfies $h(0) = 0$ and $h'(u) = (1-u)/\big(2(1+u)\big) \ge 0$ on $[0,1]$. Finally $a \ge \alpha_{\mathrm{w}}\varepsilon_H = \varepsilon_H/L_1$. □

Remark 8. The rate μ_+ depends on the curvature $|\lambda_{\min}|$ at the snapshot, which is not known when the parameters are set. All constants are therefore chosen for the worst admissible case $|\lambda_{\min}| = \varepsilon_H$, using the guaranteed bound $\log\mu_+ \ge \varepsilon_H/(2L_1)$ of Lemma 17. A snapshot with more negative curvature only escapes faster.

Remark 9 (The window rate is set by γ). The bound $\log\mu_+ \ge \varepsilon_H/(2L_1)$ is not an artefact of the analysis. By Lemma 17, $\mu_+ < 1 + a/(1-\gamma)$ with $a = \alpha_{\mathrm{w}}|\lambda_{\min}|$, so for small a the escape rate improves on the gradient-descent rate $1+a$ by the constant factor $1/(1-\gamma)$ and never by an order. At $\gamma = 1$ the characteristic polynomial $\mu^2 - (1+\gamma+a)\mu + \gamma$ instead gives $\mu_+ \approx 1 + \sqrt{a}$, the accelerated rate, but the energy (3.12) ceases to be monotone at $\gamma \ge \frac{1}{2}$, so the localization of Lemma 5 is unavailable exactly where acceleration would begin. A constant momentum bounded away from 1 therefore cannot shorten the window below $T = \Theta(L_1\phi/\varepsilon_H)$, and the ε_H^{-4} of Theorem 1 is a consequence of that restriction.

Proposition 2 (Scales)**.** *Let T be the smallest integer satisfying (3.49), let S, F, R be as in (3.50)–(3.51), and set*

$$\phi \;:=\; 1 + \log\!\left(\frac{n\,L_1\,L_2\,\Delta_f}{\varepsilon_H\,\delta}\right).$$

Then

$$T = O\Big(\frac{L_1\,\phi}{\varepsilon_H}\Big), \qquad S = \Omega\Big(\frac{\varepsilon_H}{L_2\,\phi}\Big), \qquad F = \Omega\Big(\frac{\varepsilon_H^3}{L_2^2\,\phi^3}\Big),$$

the implied constants depending only on γ, c and ω. All three hold with Θ in place of O and Ω when $|\lambda_{\min}| = \Theta(\varepsilon_H)$.

Proof. We begin by checking that the constants appearing in (3.47) are of order one. By Lemma 17, $\mu_+ \in (1,3)$, so $c_{\mathrm{esc}} = (\mu_+ - \gamma)/(\mu_+ - \mu_-) > (1-\gamma)/3$; and by Lemmas 10 and 11, c_h and $C_{\mathbf{M}}$ are bounded above by expressions involving γ alone. Hence $\Lambda = \Theta(1)$, with implied constants depending only on γ and c. This will be used throughout without further comment.

We turn to the window length, which is governed by the rate $\log\mu_+$ appearing in (3.53). Lemma 17 bounds it from below,

$$\log\mu_+ \;\ge\; \frac{\varepsilon_H}{2L_1}, \tag{3.56}$$

and it remains to control the logarithm in (3.53). By (3.54), the ratio $S(T)/R(T)$ is at most a quadratic polynomial in T whose coefficients are polynomial in L_1, L_2, ε_g and the constants above; since $\varepsilon_g \le 1$, the bracket in (3.53) is then polynomial in n, L_1, L_2, $1/\delta$ and T, so its logarithm is $O(\phi + \log T)$. Combining this with (3.56),

$$G(T) \;=\; O\Big(\frac{L_1}{\varepsilon_H}\big(\phi + \log T\big)\Big).$$

The smallest T satisfying $T \ge G(T)$ is therefore polynomially bounded in the problem data, hence $\log T = O(\phi)$ and

$$T \;=\; O\Big(\frac{L_1\,\phi}{\varepsilon_H}\Big).$$

The remaining constants follow from this bound. By (3.50), $S = \Psi/2 = \Theta\big(L_1/(L_2 T)\big)$, so an upper bound on T becomes a lower bound on S,

$$S \;=\; \Omega\Big(\frac{\varepsilon_H}{L_2\,\phi}\Big),$$

and substituting into $F = \kappa_\gamma S^2/T$, using the bound on T once more,

$$F \;=\; \Omega\bigg(L_1 \cdot \frac{\varepsilon_H^2}{L_2^2\phi^2} \cdot \frac{\varepsilon_H}{L_1\phi}\bigg) \;=\; \Omega\bigg(\frac{\varepsilon_H^3}{L_2^2\phi^3}\bigg).$$

Finally, these bounds are two-sided when the curvature at the snapshot is of order ε_H. Indeed, if $|\lambda_{\min}| = \Theta(\varepsilon_H)$ then $a = \Theta(\varepsilon_H/L_1)$, so $\log\mu_+ = \Theta(\varepsilon_H/L_1)$ by (3.55); the bound on T is then attained, and the two inequalities above become equalities in order. $\square$

3.4.3 Counting the iterations

We first fix the failure budget. Since $\bar K_{\max}$ below depends on F, which by Proposition 2 depends on ϕ, which in turn depends on $\log(1/\delta)$, we break the circularity by fixing

$$\phi_0 := 1 + \log\bigg(\frac{n\,L_1\,L_2\,\Delta_f}{\varepsilon_H\,\delta_{\rm tot}}\bigg)$$

in advance and evaluating T, S, F, R with ϕ_0 in place of ϕ. Then

$$\bar K_{\max} := \frac{\Delta_f}{(1-\omega)F} + 1, \qquad \delta := \frac{\delta_{\rm tot}}{\bar K_{\max}}, \tag{3.57}$$

are well defined, δ being the failure probability allotted to a single escape window. As $\bar K_{\max}$ is polynomial in the problem data, $\log(1/\delta) = O(\phi_0)$, so all bounds below hold with ϕ_0.

Proof of Theorem 1. Fix the parameters as in Proposition 1, with δ as in (3.57).

We first record the decrease available at each type of iteration. From Algorithm 1, a descent step has $\|g_k\| > \varepsilon_g$. If $k \in \mathcal{N}$, then (3.7) together with $\varepsilon_g \le 1$ and $1 + p - q \ge 0$ gives

$$f(x_k) - f(x_{k+1}) > c_{\mathcal{N}}\,\varepsilon_g^{\max\{1+p,\,2(1+p-q)\}},$$

while if $k \in \mathcal{R}$, then (3.8) gives $f(x_k) - f(x_{k+1}) > c_{\mathcal{R}}\,\varepsilon_g^2$. In both cases the decrease is strictly positive.

Escape windows contribute in the same way, provided they do not terminate the algorithm. Index the perturbation episodes by $j = 1, 2, \ldots$; episode j takes a snapshot $\tilde x^{(j)}$ and runs T window steps, ending at $\hat x_T^{(j)}$. If the certification (3.1) does not return at the end of episode j, then by the test itself

$$f(\tilde x^{(j)}) - f(\hat x_T^{(j)}) \;\ge\; F_{\rm cert} = (1-\omega)F > 0.$$

These decreases can now be set against the total budget. They are nonnegative, and they are attached to disjoint phases of the run: a descent step k accounts for the passage from x_k to x_{k+1}, and a non-returning episode j for the passage from $\tilde x^{(j)}$ to $\hat x_T^{(j)}$, so that no iteration belongs to two such phases. Their sum therefore telescopes to at most $f(x_0) - f_{\rm low} = \Delta_f$, therefore

$$|\mathcal{N}| \le \frac{\Delta_f}{c_{\mathcal{N}}}\,\varepsilon_g^{-\max\{1+p,\,2(1+p-q)\}}, \qquad |\mathcal{R}| \le \frac{\Delta_f}{c_{\mathcal{R}}}\,\varepsilon_g^{-2}, \qquad \bar K \le \bar K_{\max}, \tag{3.58}$$

where $\bar{K}$ is the number of episodes actually executed and $\bar{K}_{\max}$ is as in (3.57).

Termination already follows. Every iteration of Algorithm 1 lies in $\mathcal{N}$, $\mathcal{R}$ or $\mathcal{W}$, and $|\mathcal{W}| \le T\bar{K} \le T\bar{K}_{\max}$ by (3.58), so all three counts are finite and the algorithm performs finitely many iterations. Its outer loop being infinite, and exiting only through the **return** of the certification test, that test must fire; the algorithm therefore returns some $\tilde{x}$.

It remains to convert the episode count into a number of window iterations. By Proposition 2, $F = \Omega\big(\varepsilon_H^3/(L_2^2\phi_0^3)\big)$ and $T = O(L_1\phi_0/\varepsilon_H)$, so

$$\bar{K}_{\max} = O\bigg(\frac{L_2^2\Delta_f\,\phi_0^3}{\varepsilon_H^3}\bigg), \qquad |\mathcal{W}| \le T\,\bar{K}_{\max} = O\bigg(\frac{L_1L_2^2\Delta_f\,\phi_0^4}{\varepsilon_H^4}\bigg) = \widetilde{O}\bigg(\frac{L_1L_2^2\Delta_f}{\varepsilon_H^4}\bigg),$$

since ϕ_0 is logarithmic in n, $1/\delta_{\mathrm{tot}}$ and the problem constants.

Everything so far has been deterministic. Randomness enters only in establishing that the point returned by the certification test really is second-order stationary. For each episode j let

$$\mathcal{E}_j \coloneqq \big\{\lambda_{\min}(\nabla^2 f(\tilde{x}^{(j)})) \le -\varepsilon_H \ \text{ and the certification returns } \tilde{x}^{(j)}\big\}$$

be the event that episode j returns a point which is in fact a strict saddle. On $\{\lambda_{\min}(\nabla^2 f(\tilde{x}^{(j)})) \le -\varepsilon_H\}$ the snapshot satisfies the hypotheses of Lemma 15, and returning means $f(\tilde{x}^{(j)}) - f(\hat{x}_T^{(j)}) < F_{\mathrm{cert}}$, which Lemma 15 rules out except on an event of probability at most δ; hence $\mathbb{P}(\mathcal{E}_j) \le \delta$. Since at most $\bar{K}_{\max}$ episodes occur, the union bound and (3.57) give

$$\mathbb{P}\bigg(\bigcup_{j\le\bar{K}_{\max}} \mathcal{E}_j\bigg) \;\le\; \bar{K}_{\max}\,\delta \;=\; \delta_{\mathrm{tot}}.$$

We conclude on the complement of $\bigcup_j \mathcal{E}_j$, an event of probability at least $1-\delta_{\mathrm{tot}}$. There, the returned point $\tilde{x}$ satisfies $\|\nabla f(\tilde{x})\| \le \varepsilon_g$, since it is a snapshot, and $\lambda_{\min}(\nabla^2 f(\tilde{x})) > -\varepsilon_H$, since otherwise some $\mathcal{E}_j$ would have occurred; it is therefore an $(\varepsilon_g,\varepsilon_H)$-SOSP. Summing the three counts of (3.58) with the bound on $|\mathcal{W}|$ obtained above,

$$N = |\mathcal{N}| + |\mathcal{R}| + |\mathcal{W}| \;\le\; \frac{\Delta_f}{c_{\mathcal{R}}}\,\varepsilon_g^{-2} + \frac{\Delta_f}{c_{\mathcal{N}}}\,\varepsilon_g^{-\max\{1+p,\,2(1+p-q)\}} + \widetilde{O}\bigg(\frac{L_1L_2^2\Delta_f}{\varepsilon_H^4}\bigg),$$

which is the bound of Theorem 1. $\square$

Chapter 4

Numerical Experiments

In this chapter, we show the numerical experiments realized to complement the theory of previous sections. Four things are tested, in order. Section 4.2 resolves the constraint system and shows that the certified constants are not executable, then we propose a solution for implementation. Section 4.3 isolates the descent phase, where the restart exponents (p, q) act. Section 4.4 measures the escape against PGD [27] and PAGD [28], in iterations and in oracle evaluations. Section 4.5 varies the design choices of Section 3.3, that is, the window length, the window subroutine, and the use of a negative-curvature oracle in its place to identify which of them bind. Section 4.6 collects the caveats.

4.1 Experimental Design

We use two strict-saddle instances, both with closed-form minimizers and analytically certified constants. The first is a quadratic saddle with quartic confinement,

$$f(x) = \tfrac{1}{2}x^\top Hx + \tfrac{\nu}{4}\|x\|^4, \qquad H = \mathrm{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_1 = -\lambda_- < 0, \tag{4.1}$$

with minimizers $x^\star = \pm\sqrt{\lambda_-/\nu}\, e_1$ and $f^\star = f_{\mathrm{low}} = -\lambda_-^2/(4\nu)$. Its spectrum is an input, so ε_H, μ_+ and the confluence point $\lambda^* = L_1(1-\sqrt{\gamma})^2$ of Section 3.3.3 are known exactly. We take $n = 20$, $\lambda_- = 1/2$, $\nu = 1/10$ and $\lambda_2, \dots, \lambda_n \sim \mathrm{Unif}[\frac{1}{2}, 1]$, redrawn per repetition.

The second is symmetric low-rank matrix factorization,

$$f(U) = \tfrac{1}{4}\|UU^\top - M\|_F^2, \qquad M = U^\star(U^\star)^\top \in \mathbb{R}^{d\times d} \text{ of rank } r, \tag{4.2}$$

with $n = dr$, every local minimizer global ($f^\star = f_{\mathrm{low}} = 0$) and all saddles strict [18, 19]. Write $\sigma_1 \geq \cdots \geq \sigma_r > 0$ for the nonzero eigenvalues of M; the origin is then a strict saddle with $\lambda_{\min} = -\sigma_1$. Planting $U^\star = Q\,\mathrm{diag}(\sqrt{\sigma_1}, \dots, \sqrt{\sigma_r})$ with Q an orthonormal frame gives M the spectrum we want. We take $d = 10$, $r = 7$ and $\sigma_1, \dots, \sigma_r$ evenly spaced on $[1, 3]$ for numerics.

Both objectives are quartic, so global Lipschitz constants do not exist and Assumptions 1 and 2 are certified on a working ball $\|x\| \leq B_{\mathrm{dom}}$, taken as twice the norm of the known minimizers. This gives $L_1 = \max_i|\lambda_i| + 3\nu B_{\mathrm{dom}}^2$ and $L_2 = 6\nu B_{\mathrm{dom}}$ for (4.1), and $L_1 = 3B_{\mathrm{dom}}^2 + \|M\|_2$, $L_2 = 6B_{\mathrm{dom}}$ for (4.2). The resulting Lipschitz constants are conservative, which inflates the window length T; Table 4.1 quantifies this.

Starting points The two groups of experiments use different starts.

For the descent-phase experiments of Section 4.3, starts are generic and the perturbation mechanism is switched off, so that the measured quantity is the descent iteration alone.

For the escape experiments of Sections 4.4 and 4.5, starts lie on the saddle's stable manifold, so that the escape mechanism is actually exercised: from a generic initialization a strict saddle is avoided almost surely, and no perturbation would ever be triggered. For (4.1), x_0 is a small Gaussian with its negative-eigenspace component zeroed. For (4.2), the columns of U_0 lie in the orthogonal complement of range(M), which requires $d > r$; then $MU_0 = 0$, so $\|\nabla f(U_0)\| = O(\|U_0\|^3)$ is negligible while $\lambda_{\min} \approx -\sigma_1(M)$. These experiments therefore measure the escape mechanism *conditional on meeting a saddle*, and say nothing about how often that happens.

Methods and parameters All methods share the stepsize $\alpha_{\mathrm{w}} = 1/L_1$ and, where applicable, the escape budget $(T, R, F_{\mathrm{cert}})$; no parameter is tuned per method or per instance. Besides gradient descent we run PGD [27] and PAGD [28], both given the same escape budget; for PAGD we keep $\eta_{\mathrm{PAGD}} = 1/L_1$ and rescale $\theta_{\mathrm{PAGD}} = \min\{\frac{1}{2}, 1/(2\sqrt{L_1/\varepsilon_H})\}$ so that its exploitation threshold matches the reference configuration. We run Algorithm 1 with the PRP+ beta, and Algorithm 1 with the perturbations disabled, referred to below as *safeguarded NCG*.

We also report an optional variant of Algorithm 1, the *early exit*. Since the F_{cert} decrease is deterministic, a window may be left at the first step satisfying

$$f(x_k) - f(\tilde{x}) \le -F_{\mathrm{cert}} \quad \text{and} \quad \|g_k\| > \varepsilon_g, \tag{4.3}$$

handing control back to the descent. A *returned* SOSP still requires a full *failing* window, so the certificate is untouched and only successful escapes are shortened. Both conditions are needed: F_{cert} is a certification threshold, not a per-step progress signal, so exiting on the decrease alone would return while still at $\|g\| \le \varepsilon_g$ and re-perturb at once, giving a cycle of length-one windows.

The restart test (3.5) is run with

$$\sigma = 10^{-2}, \qquad \kappa = 10^2, \qquad p = \tfrac{1}{2}, \qquad q = 1, \tag{4.4}$$

and the Armijo search with $\eta = 10^{-4}$, backtracking factor $\theta = 1/2$ and at most 200 backtracks. The exponents place the descent bound $\varepsilon_g^{-\max\{1+p,\, 2(1+p-q)\}}$ of Theorem 1 at $\varepsilon_g^{-3/2}$, inside the regime where it improves on the ε_g^{-2} of a restart step; the natural-looking $p = q = 1$ would give exactly 2 and no improvement. The scales σ and κ keep the safeguard off the binding path: with $q > p$, Lemma 3 places a gradient floor at $(\sigma/\kappa)^{1/(q-p)} = 10^{-8}$, and $d = -g$ passes the first condition of (3.5) whenever $\|g\| \ge \sigma^{1/(1-p)} = 10^{-4}$, both below $\varepsilon_g = 10^{-3}$. Across every run reported here the test never fires, so all recorded descent steps are accepted conjugate-gradient steps.

The remaining tolerances are $\varepsilon_g = 10^{-3}$, $\delta_{\mathrm{tot}} = 0.1$, $\Delta_f = f(x_0) - f_{\mathrm{low}}$ and $\gamma = 0.45$, just below the $\gamma < \frac{1}{2}$ required by Lemma 4. The curvature tolerance is oracle-derived,

$$\varepsilon_H = 0.8\,|\lambda_{\min}(\nabla^2 f(\mathrm{saddle}))|, \tag{4.5}$$

the factor 0.8 ensuring that the planted saddle is strict at the declared tolerance, so that a return there would be a genuine false certification. A practitioner does not have this quantity; we return to the point in Section 4.6.

Protocol Every reported number is a median over 20 independent repetitions, each with a fresh instance, starting point and perturbation draw. Runs are capped at $3 \cdot 10^4$ iterations.

We distinguish throughout between *reaching* an SOSP and *returning* one. Reaching is the index of the first $(\varepsilon_g, \varepsilon_H)$-SOSP along the trajectory, certified offline with the true Hessian; returning is the iteration at which the algorithm itself stops and certifies. The two differ because certification requires a full failing window, and the gap between them is studied in Section 4.4.

4.2 The parameter ledger

The resolved system (C1)–(C4) is used to *verify* that the constraints are solvable, not to run the algorithms. At the reference configuration for the quadratic saddle problem, ($n = 20$, $L_1 = 7$,

$L_2 = 2.7$, $\varepsilon_g = 10^{-3}$, $\varepsilon_H = 0.4$, $\gamma = 0.45$) it returns

$$T = 636, \qquad R \approx 2 \cdot 10^{-14}, \qquad F \approx 9 \cdot 10^{-17}, \qquad \bar{K} \approx 2 \cdot 10^{16} :$$

a perturbation radius at the scale of double-precision round-off and a certification threshold below it. Executed literally, such a ledger measures round-off rather than dynamics. The runs therefore use the Θ-scales of Proposition 2 with the polylogarithmic factors set to one,

$$\alpha_{\mathrm{w}} = \frac{1}{L_1}, \quad S = \frac{\varepsilon_H}{L_2}, \quad F = \frac{L_1(\frac{1}{2} - \gamma)S^2}{T}, \quad R = \min\left\{\frac{\omega F}{2\varepsilon_g}, \sqrt{\frac{\omega F}{L_1}}, S\right\}, \quad F_{\mathrm{cert}} = (1-\omega)F, \tag{4.6}$$

with $\omega = \frac{1}{2}$ and T resolved from the fixed point of (C4), so that the escape probability remains honest. The single constraint we drop is (C2), the forcing budget, whose constant Λ carries the accumulated worst-case slack of the induction. Hence, using these practical values, *the runs below are not certified by Theorem 1*. They test whether the mechanism the analysis describes behaves as predicted at executable scales.

instance	n	L_1	L_2	$\lvert\lambda_{\min}\rvert$	ε_H	L_1/ε_H	T	F_{cert}
quadratic saddle	20	6.97	2.68	0.50	0.40	17	208	$1.9 \cdot 10^{-5}$
matrix factorization	70	171	44.9	3.00	2.40	71	952	$1.3 \cdot 10^{-5}$

Table 4.1: Practical ledger per instance. The window length T tracks L_1/ε_H, and that ratio is dominated by the conservatism of the ball-certified L_1 rather than by the geometry of the instance.

4.3 The descent phase

The safeguarded descent is far more efficient than a gradient step. The iterations to $\|g\| \le \varepsilon_g$ of the descent phase is reported in Table 4.2.

	quadratic saddle				matrix factorization			
ε_g	10^{-2}	10^{-4}	10^{-6}	10^{-8}	10^{-2}	10^{-4}	10^{-6}	10^{-8}
gradient descent	109	144	174	204	1594	1989	2469	3234
safeguarded NCG, FR	4	17	21	27	83	102	108	117
safeguarded NCG, PRP+	4	15	20	**26**	20	26	35	**47**

Table 4.2: Iterations to $\|g\| \le \varepsilon_g$ of gradient descent, safeguarded NCG using the FR beta, and safeguarded NCG using the PRP+ beta. Safeguarded NCG with PRP+ converges the fastest to the ε_g-FOSP.

We can derive three interesting observations. First, the restart test (3.5) never fires: with the exponents of (4.4) every recorded step is an accepted conjugate step, so what is measured is the conjugate direction itself and not a disguised gradient descent. Second, this is the only place in the section where we test the two betas against each other. PRP+ works better (47 against 117) so we stick to it for the rest of the section. Third, a caveat on what the table does *not* show. The bound of Theorem 1 predicts a difference in *rate*, $\varepsilon_g^{-3/2}$ against ε_g^{-2}, so the ratio between the two counts ought to grow without bound as ε_g falls. It does not: across the four tolerances that ratio is $27, 10, 9, 8$ on the quadratic saddle and $80, 77, 71, 69$ on matrix factorization, which is flat or shrinking. Neither method comes close to its worst-case rate here: gradient descent needs barely twice as many iterations at $\varepsilon_g = 10^{-8}$ as at 10^{-2}, whereas an ε_g^{-2} rate over that range would demand a factor 10^{12}. Both bounds are therefore far from tight on these instances. What the table reports is therefore a large constant-factor advantage of the conjugate direction, not the asymptotic separation that the exponents (4.4) were chosen to permit; exhibiting the latter would require an instance on which gradient descent actually attains its ε_g^{-2} worst case.

4.4 The escape phase

	quadratic saddle			matrix factorization		
method	reach	evals	return	reach	evals	return
gradient descent	—	—	n/a	6248	6249	n/a
safeguarded NCG	—	—	n/a	—	—	n/a
PGD	210	213	450	2181	2187	3119
PAGD	149	596	444	570	2280	1904
Algorithm 1	99	105	422	956	967	1909
Algorithm 1, early exit	**50**	**68**	**256**	**137**	**203**	**1087**

Table 4.3: Cost to *reach* the first $(\varepsilon_g, \varepsilon_H)$-SOSP (iterations, and gradient plus function evaluations) and to *return* one. Medians over 20 repetitions. "—": no SOSP reached within the cap on any repetition. "n/a": no certification mechanism, gradient descent never certifies, and safeguarded NCG halts on $\|g\| \leq \varepsilon_g$ after two to five iterations, at the saddle.

Both unperturbed methods either do not succeed or pay a high price to do so. With the perturbations disabled the safeguarded descent never reaches an SOSP. It stops after two to five iterations at the strict saddle it was started on, where $\|g\| \leq \varepsilon_g$ already holds. A first-order stop is not a second-order certificate, and that gap is what the escape window fills. Gradient descent escapes the matrix factorization saddle, but pays 6248 iterations for it.

All three perturbed methods escape, and how they compare depends on whether we count reaching or returning.

In *reaching*, Algorithm 1 wins on the quadratic saddle (99 against PAGD's 149) but loses on matrix factorization (956 against 570). The reason is the window length rather than the descent. On matrix factorization the window opens at iteration 0 and the decrease that certifies the escape arrives after a median of 113 of its 952 steps, but the first iterate that is genuinely an SOSP appears only at 956, just after the window closes. Everything in between is spent inside a window that has already done its job. On the quadratic saddle the corresponding numbers are 33.5 of 208, against a reach of 99. Leaving the window as soon as (4.3) certifies the escape removes precisely those steps: the reach falls to 137, and to 50 from 99 on the quadratic saddle.

In *returning*, the picture is different, because every perturbed method must first spend one full window that *fails*. That cost is the same for all of them, so the differences almost disappear: PAGD returns at 1904 and Algorithm 1 at 1909 on matrix factorization, a gap of 0.3%, where their reaches differed by a factor 1.7. The early exit only shortens windows that succeed, so it improves returning by a factor 1.6–1.8, against the 2 to 7.0 it gives in reaching.

The evaluation counts tell a second story, visible in Figure 4.1. One PAGD iteration costs four evaluations, two gradient and two function, so wherever PAGD leads in iterations, it trails in evaluations. For example, it has 2280 evaluations against 967 for Algorithm 1 on matrix factorization, even before the early exit. Algorithm 1 spends one gradient per window step and none on the restart test, and the Armijo line search does spend extra function values; but the runs here are dominated by window steps, so few line searches are performed and the total stays close to one evaluation per iteration (1.01 and 1.07). The early exit hands control back to the descent sooner and therefore buys more line searches, which is where the line-search cost becomes visible (1.37 and 1.48), still far below PAGD.

Figure 4.2 states the same comparison without reference to a median: for every budget K, the fraction of repetitions that have delivered a certified SOSP within K. A curve lying to the left is better *at every budget*, so the ranking does not depend on where one chooses to stop. The early-exit variant is leftmost in all four panels, and the second row, the same profile counted in oracle evaluations, moves PAGD from second-best to worst on the quadratic saddle.

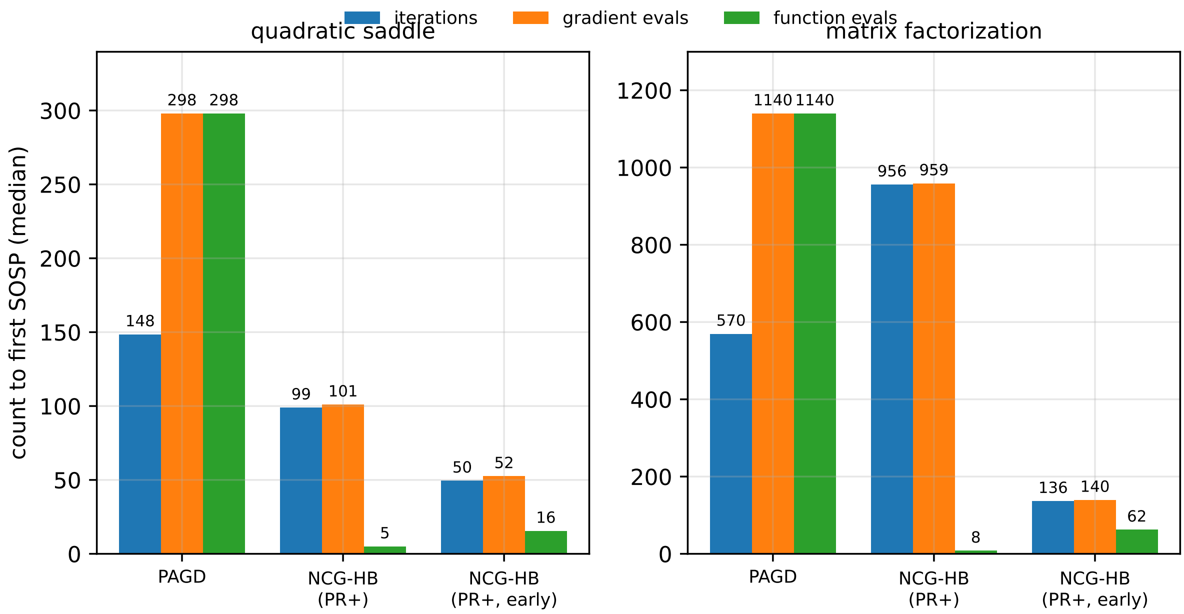

Figure 4.1: Iterations, gradient evaluations and function evaluations to the first SOSP, medians over 20 repetitions. The figure separates the two currencies: PAGD's function evaluations match its gradient evaluations one for one, while those of Algorithm 1 — spent only in the line search of the descent phase — remain two orders of magnitude smaller.

4.5 Which design choices bind

Window length The window length T given by the ledger is much longer than the escape needs, so it is natural to ask how far it can be cut before the method starts returning points that are not SOSPs. Figure 4.3 replaces T by τT for $\tau = 1/16, \ldots, 1$ and counts, over 20 repetitions, how often the returned point is a genuine SOSP.

At $\tau = 1/16$ every repetition returns a strict saddle, on both instances. As τ grows the method becomes correct again, but not at the same point for the two problems: matrix factorization is correct 16 times out of 20 at $\tau = 1/8$ and on all 20 repetitions at $\tau = 1/4$, while the quadratic saddle is correct only 4 times out of 20 at $\tau = 1/8$ and 17 out of 20 at $\tau = 1/4$, and needs $\tau = 1/2$ to be reliable. Two things follow. The window can safely be shortened by a factor of two to four, so most of the certified budget is unused; but the safe factor is not the same on the two instances, and below it the method fails far more often than the 10% the ledger allows for (15% at $\tau = 1/4$ on the quadratic saddle, 20% at $\tau = 1/8$ on matrix factorization). Since shortening the window is exactly what removing the logarithmic factor in (C4) would do, this also shows that the factor is not decorative.

Shortening the window to the smallest safe value saves a good deal of work: the run then certifies and stops after 214 iterations instead of 422 on the quadratic saddle ($\tau = 1/2$), and 504 instead of 1909 on matrix factorization ($\tau = 1/4$); with the early exit as well, 152 and 372. We report this to show how much slack the ledger carries, not as something a user could do: finding the smallest safe τ requires knowing when a returned point is a strict saddle, which requires the Hessian, the same objection that applies to ε_H in (4.5). Shortening T while leaving F_{cert} at its ledger value also breaks the identity (C1) that relates the two.

Window subroutine Figure 4.4 runs the window alone from the same perturbed snapshot with the three subroutines of Remark 3. The escape-time distributions satisfy HB $\approx$ AGD $<$ GD on both instances (medians 33.5/32.0/53.0 and 113/112/198), with no failures. At the constant

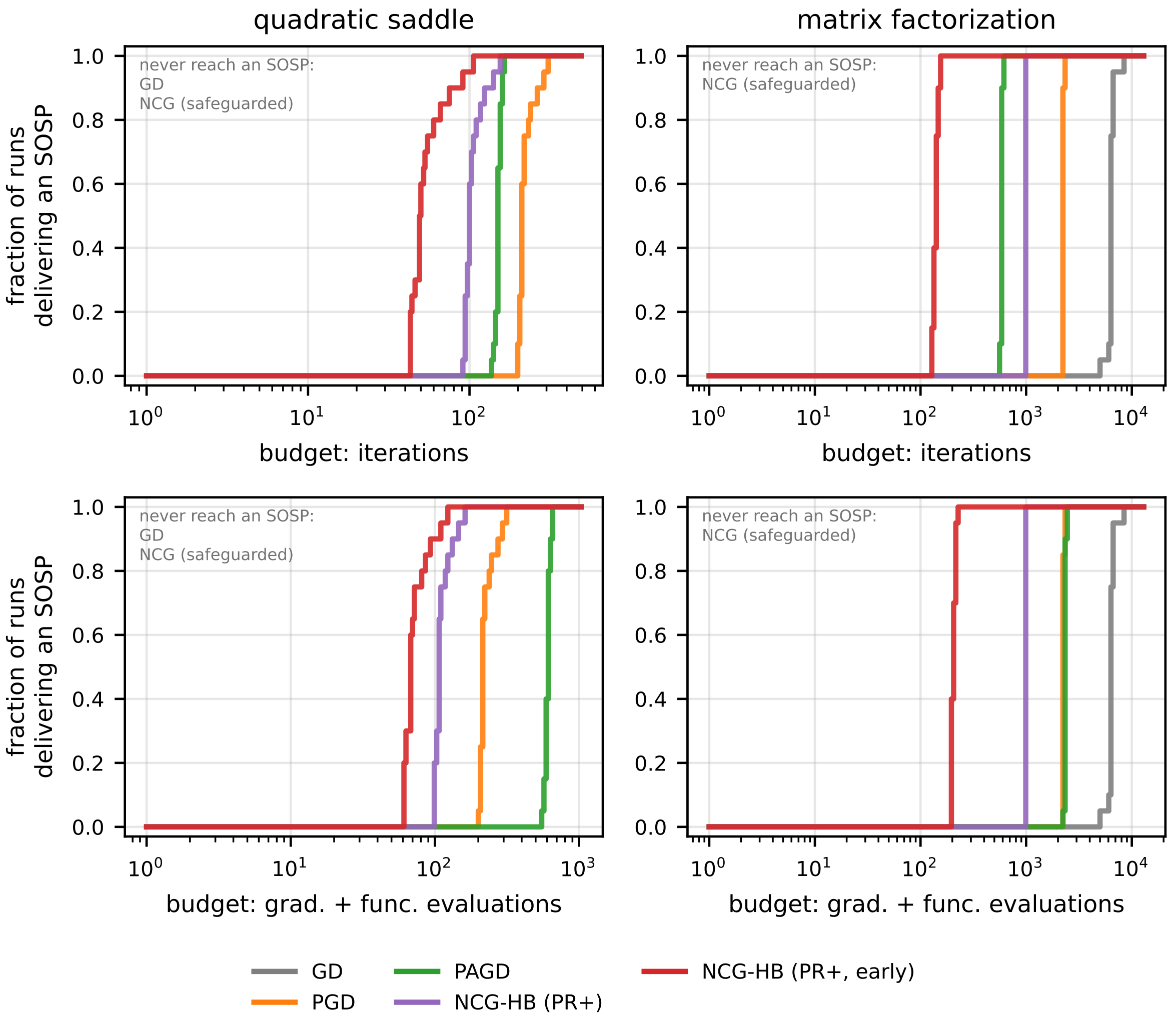

Figure 4.2: Fraction of the 20 repetitions that have reached an $(\varepsilon_g, \varepsilon_H)$-SOSP within a given budget of iterations (top) and of oracle evaluations (bottom). Methods absent from a panel never reach one on that instance and are named in the corner.

momentum $\gamma < \frac{1}{2}$ that the localization energy requires, the accelerated window ties HB and neither changes the order relative to plain gradient. This is the empirical content of Remark 9, and the reason the window is left as HB everywhere else.

Against a negative-curvature oracle Replacing the window by an explicit query of the smallest Hessian eigenpair, that is building an NCG/NCD hybrid, escapes each saddle in a single step: 11 outer iterations against 99 on the quadratic saddle and 23 against 956 on matrix factorization, using two and three eigendecompositions per run.

This is the exchange Algorithm 1 makes. As Section 2.3 records, the oracle is invoked at every candidate stationary point, and even in its cheapest form each invocation costs several Hessian-vector products or gradient differences. The HB window attains the same $(\varepsilon_g, \varepsilon_H)$ guarantee with no second-order access of any kind, and pays for it in outer iterations.

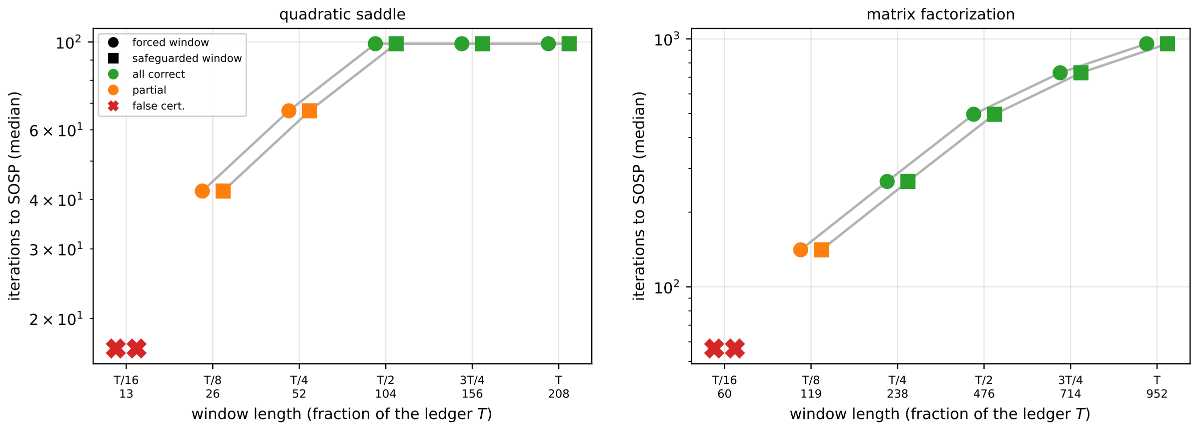

Figure 4.3: Iterations to the SOSP as the window budget is cut to a fraction τ of each instance's ledger T (208 and 952; the absolute window length is given under each tick), over 20 repetitions. Green: every repetition certifies a true SOSP; orange: some false certifications; red cross: every repetition false-certifies.

4.6 Limitations

Four caveats bound what the above establishes. *(i)* The runs use the practical ledger of (4.6) with (C2) dropped, so they are not certified by Theorem 1. *(ii)* The curvature tolerance (4.5) is derived from the known saddle curvature of each instance; a practitioner has neither that nor the smallest reliable window fraction of Section 4.5, and the agnostic coupling $\varepsilon_H = \sqrt{L_2 \varepsilon_g}$, combined with the ball-certified L_1, inflates T past any practical budget. *(iii)* The escape experiments start on the stable manifold, so they measure the mechanism conditional on meeting a saddle and say nothing about how often that occurs. *(iv)* On those starts the runs are almost entirely window steps (a median of four descent steps against 418 and 1904 window steps) so Table 4.3 compares escape behaviour, and the descent-phase evidence rests on Section 4.3 alone. Two instances, both with planted structure, are also a narrow base from which to generalize.

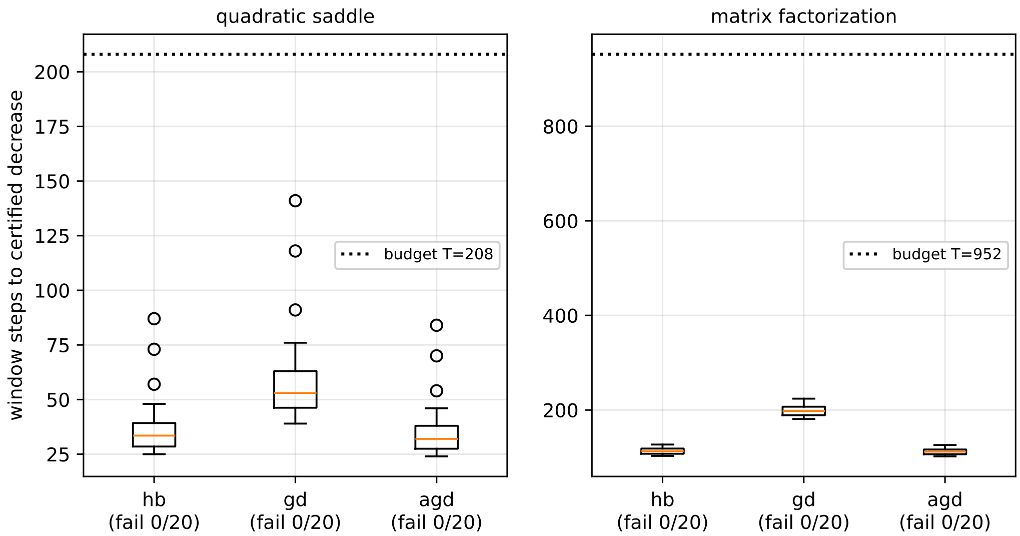

Figure 4.4: Escape time of a single window, by subroutine, over 20 repetitions per instance, at $\gamma = 0.45$ and perturbation radius $R = 1.6 \cdot 10^{-3}$ (quadratic saddle) and $1.8 \cdot 10^{-4}$ (matrix factorization). Dotted line: the window budget T.

Chapter 5

Local Convergence under the Strict Saddle Property

So far the target has been an $(\varepsilon_g, \varepsilon_H)$-SOSP, the standard objective in nonconvex optimization. Without further structure nothing forces it to be a local minimizer. In this chapter we add one landscape assumption, the strict saddle property of [18], under which a certified SOSP is guaranteed to lie near a local minimizer, in a region where f is strongly convex. A safeguarded NCG phase run from that point then converges to the minimizer, with an explicit rate.

Section 5.1 shows that under the strict saddle property the point returned by Algorithm 1 lies in a strongly convex basin. Section 5.2 defines the local phase (Algorithm 2), which continues from that point with safeguarded NCG. Section 5.3 counts its iterations: restart steps contract linearly, while non-restart steps obey a recursion whose behaviour is linear, finite or sublinear according to the exponent $r = \frac{1}{2}\max\{1+p,\, 2(1+p-q)\}$ set by the restart parameters of (3.5). Section 5.4 sums the three counts.

5.1 Setting

Definition 1 (Strict Saddle Property, [18])**.** f is $(\zeta_{\mathrm{ss}}, \beta_{\mathrm{ss}}, \gamma_{\mathrm{ss}}, \delta_{\mathrm{ss}})$-strict saddle if for every x, at least one of the following holds:

1. **Large gradient** $\|\nabla f(x)\| \geq \zeta_{\mathrm{ss}}$,
2. **Negative curvature** $\lambda_{\min}(\nabla^2 f(x)) \leq -\beta_{\mathrm{ss}}$,
3. **Near a local minimizer** there is a local minimizer $x^\star$ with $\|x - x^\star\| \leq \delta_{\mathrm{ss}}$ and f is γ_{ss}-strongly convex on $B_{x^\star}(2\delta_{\mathrm{ss}})$.

The three cases of Definition 1 are exhaustive: every point satisfies at least one. The point of running Algorithm 1 with tolerances below the strict-saddle thresholds is that it rules out the first two, leaving the third.

Lemma 18 (Entry into the strongly convex basin)**.** *Let f be $(\zeta_{\mathrm{ss}}, \beta_{\mathrm{ss}}, \gamma_{\mathrm{ss}}, \delta_{\mathrm{ss}})$-strict saddle and run Algorithm 1 with $\varepsilon_g < \zeta_{\mathrm{ss}}$ and $\varepsilon_H < \beta_{\mathrm{ss}}$. On the success event of Theorem 1, which has probability at least $1 - \delta_{\mathrm{tot}}$, the returned point $\tilde{x}$ lies within δ_{ss} of a local minimizer $x^\star$ on whose neighbourhood $B_{x^\star}(2\delta_{\mathrm{ss}})$ the function f is γ_{ss}-strongly convex.*

Proof. On the success event, Theorem 1 gives $\|\nabla f(\tilde{x})\| \leq \varepsilon_g$ and $\lambda_{\min}(\nabla^2 f(\tilde{x})) \geq -\varepsilon_H$. Since $\varepsilon_g < \zeta_{\mathrm{ss}}$, case (1) of Definition 1 fails at $\tilde{x}$, and since $\varepsilon_H < \beta_{\mathrm{ss}}$, so does case (2). The three cases being exhaustive, case (3) holds at $\tilde{x}$. □

We write k_0 for the iteration at which the local phase begins, so that $x_{k_0} = \tilde{x}$, and $\varepsilon_{\mathrm{loc}} \in (0, \varepsilon_g)$ for the target accuracy of that phase.

Write

$$\Delta_k \;\coloneqq\; f(x_k) - f(x^\star). \tag{5.1}$$

For $x_k \in B_{x^\star}(2\delta_{\mathrm{ss}})$, smoothness and γ_{ss}-strong convexity give the standard sandwich (using $\nabla f(x^\star) = 0$)

$$\begin{aligned} \frac{\gamma_{\mathrm{ss}}}{2}\|x_k - x^\star\|^2 \;\le\; \Delta_k \;\le\; \frac{L_1}{2}\|x_k - x^\star\|^2, \\ 2\gamma_{\mathrm{ss}}\,\Delta_k \;\le\; \|g_k\|^2 \;\le\; 2L_1\,\Delta_k, \end{aligned} \tag{5.2}$$

the left half of the second line being the Polyak–Łojasiewicz inequality, which we use repeatedly below.

At entry, $\|x_{k_0} - x^\star\| \le \delta_{\mathrm{ss}}$ and (5.2) give

$$\Delta_{k_0} \;\le\; \frac{L_1\delta_{\mathrm{ss}}^2}{2}. \tag{5.3}$$

Since Δ_k is nonincreasing along the local phase (Lemmas 1 and 2), the right half of the second line of (5.2) together with (5.3) gives $\|g_k\|^2 \le 2L_1\Delta_k \le 2L_1\Delta_{k_0} \le L_1^2\delta_{\mathrm{ss}}^2$, that is

$$\|g_k\| \;\le\; L_1\,\delta_{\mathrm{ss}} \qquad \text{for all } k \ge k_0 \text{ with } x_k \in B_{x^\star}(2\delta_{\mathrm{ss}}). \tag{5.4}$$

5.2 The Local Phase

Lemma 18 places the certified point inside a strongly convex basin. The local phase simply continues from there with safeguarded NCG, and the run stops when the gradient falls below $\varepsilon_{\mathrm{loc}}$.

Algorithm 2 PRNCG-HB-L: Algorithm 1 followed by local refinement

Input: the inputs of Algorithm 1, with $\varepsilon_g < \zeta_{\mathrm{ss}}$ and $\varepsilon_H < \beta_{\mathrm{ss}}$; local tolerance $\varepsilon_{\mathrm{loc}} \in (0, \varepsilon_g)$.
Run Algorithm 1; let $\tilde{x}$ be the point it returns.
$x_{k_0} \leftarrow \tilde{x}$; $g_{k_0} \leftarrow \nabla f(x_{k_0})$; $d_{k_0} \leftarrow -g_{k_0}$ ▷ fresh direction
for $k = k_0, k_0 + 1, \ldots$ **do**
    if $\|g_k\| \le \varepsilon_{\mathrm{loc}}$ **then**
        return x_k ▷ Theorem 2
    end if
    $\alpha_k \leftarrow \theta^{j_k}$, j_k minimal with $f(x_k + \theta^{j_k} d_k) \le f(x_k) + \eta\,\theta^{j_k} g_k^\top d_k$
    $x_{k+1} \leftarrow x_k + \alpha_k d_k$; $g_{k+1} \leftarrow \nabla f(x_{k+1})$
    $d_{k+1}^{\mathrm{trial}} \leftarrow -g_{k+1} + \beta_{k+1} d_k$
    if (3.5) holds for d_{k+1}^{trial} **then**
        $d_{k+1} \leftarrow d_{k+1}^{\mathrm{trial}}$
    else
        $d_{k+1} \leftarrow -g_{k+1}$
    end if
end for

The analysis below targets $\Delta_k \le \varepsilon_{\mathrm{loc}}^2/(2L_1)$, which by the right half of (5.2) implies the algorithmic stopping test $\|g_k\| \le \varepsilon_{\mathrm{loc}}$; bounding the number of iterations to reach the former therefore bounds the running time of Algorithm 2.

For the rest of the chapter we abbreviate

$$a_p \coloneqq 1 + p, \qquad b_p \coloneqq 2(1 + p - q), \qquad r \coloneqq \frac{\max\{a_p, b_p\}}{2}, \tag{5.5}$$

where p and q are the restart parameters of (3.5).

The local iterations are of two kinds. For $k \in \mathcal{R}$, combining (3.8) with the Polyak–Łojasiewicz inequality of (5.2) turns the per-step decrease into a contraction,

$$\Delta_{k+1} \;\le\; (1 - 2\gamma_{\rm ss} c_{\mathcal{R}})\, \Delta_k, \tag{5.6}$$

so restart steps converge linearly on their own. For $k \in \mathcal{N}$, (3.7) and (5.5) give only

$$\Delta_k - \Delta_{k+1} \;\ge\; c_{\mathcal{N}} \min\left\{\|g_k\|^{a_p}, \|g_k\|^{b_p}\right\},$$

whose behaviour depends on the exponents and is analysed in Section 5.3.

5.3 Iteration counts

Restart steps are counted directly from the contraction of (5.6). Non-restart steps need one preliminary observation: the decrease guarantee (3.7) involves $\min\{\|g_k\|^{a_p}, \|g_k\|^{b_p}\}$, and which of the two exponents is active depends on whether the gradient exceeds one. Since $t \mapsto t^s$ is increasing in s for $t \ge 1$ and decreasing in s for $0 < t \le 1$,

$$\min\left(\|g_k\|^{a_p}, \|g_k\|^{b_p}\right) = \begin{cases} \|g_k\|^{\min(a_p, b_p)} & \text{if } \|g_k\| \ge 1, \\ \|g_k\|^{\max(a_p, b_p)} = \|g_k\|^{2r} & \text{if } \|g_k\| \le 1. \end{cases} \tag{5.7}$$

We therefore count non-restart steps in two groups, according to whether $\|g_k\| \ge 1$ or $\|g_k\| \le 1$, writing $K^H_{\rm loc}$ and $K^L_{\rm loc}$ for the two counts. The first group is normally empty, as the following remark shows.

Remark 10 (Large-gradient steps are rare). The local phase is entered at the certified point $\tilde{x}$, where $\|\nabla f(\tilde{x})\| \le \varepsilon_g$. The Polyak–Łojasiewicz inequality (5.2) gives the entry bound $\Delta_{k_0} \le \varepsilon_g^2/(2\gamma_{\rm ss})$, sharper than the positional bound (5.3) whenever $\varepsilon_g \le \sqrt{\gamma_{\rm ss} L_1}\, \delta_{\rm ss}$. Since Δ_k is nonincreasing, smoothness then gives $\|g_k\| \le \sqrt{2L_1 \Delta_{k_0}} \le \varepsilon_g \sqrt{L_1/\gamma_{\rm ss}}$ at every local-phase iterate. Consequently, if $\varepsilon_g \le \sqrt{\gamma_{\rm ss}/L_1}$, no local iterate has $\|g_k\| \ge 1$ and $K^H_{\rm loc} = 0$. We state the bound in full nonetheless, since ε_g is a free tolerance and the condition need not hold.

5.3.1 Restart steps

Lemma 19 (Restart steps in the local phase)**.** *Suppose $x_k \in B_{x^\star}(2\delta_{\rm ss})$ for every local-phase iterate and $2\gamma_{\rm ss} c_{\mathcal{R}} < 1$. Then the number of restart steps performed before $\Delta_k \le \varepsilon^2_{\rm loc}/(2L_1)$ is reached satisfies*

$$K^{\mathcal{R}}_{\rm loc} \;\le\; \left\lceil \frac{1}{2\gamma_{\rm ss} c_{\mathcal{R}}} \log\left(\frac{L_1^2 \delta_{\rm ss}^2}{\varepsilon^2_{\rm loc}}\right) \right\rceil. \tag{5.8}$$

Proof. Write $\theta_{\mathcal{R}} := 1 - 2\gamma_{\rm ss} c_{\mathcal{R}} \in (0,1)$ and let $k_0 \le \tau_1 < \tau_2 < \dots$ be the restart steps of the local phase. By (5.6), $\Delta_{\tau_i+1} \le \theta_{\mathcal{R}} \Delta_{\tau_i}$, and every other local-phase step decreases Δ_k as well (Lemmas 1 and 2), so

$$\Delta_{\tau_K+1} \;\le\; \theta_{\mathcal{R}}^K\, \Delta_{k_0}.$$

Suppose the stopping condition has not been met at step $\tau_K + 1$, that is $\Delta_{\tau_K+1} > \varepsilon^2_{\rm loc}/(2L_1)$. Then $\theta^K_{\mathcal{R}} \Delta_{k_0} > \varepsilon^2_{\rm loc}/(2L_1)$, and taking logarithms,

$$K \log \theta_{\mathcal{R}} \;>\; \log\left(\frac{\varepsilon^2_{\rm loc}}{2L_1 \Delta_{k_0}}\right).$$

Dividing by $\log\theta_{\mathcal{R}} < 0$ reverses the inequality,

$$K \;<\; \frac{1}{-\log\theta_{\mathcal{R}}} \log\left(\frac{2L_1 \Delta_{k_0}}{\varepsilon^2_{\rm loc}}\right),$$

and since $-\log(1-x) \geq x$ on $[0,1)$ gives $1/(-\log\theta_{\mathcal{R}}) \leq 1/(2\gamma_{\rm ss}c_{\mathcal{R}})$, the entry bound (5.3) yields

$$K \;<\; \frac{1}{2\gamma_{\rm ss}c_{\mathcal{R}}}\,\log\!\left(\frac{2L_1\Delta_{k_0}}{\varepsilon_{\rm loc}^2}\right) \;\leq\; \frac{1}{2\gamma_{\rm ss}c_{\mathcal{R}}}\,\log\!\left(\frac{L_1^2\delta_{\rm ss}^2}{\varepsilon_{\rm loc}^2}\right).$$

Hence the number of restart steps before the stopping condition is met is at most the ceiling of the right-hand side, which is (5.8). □

5.3.2 Large-gradient steps

Lemma 20 (Count of large-gradient steps)**.** *Suppose $x_k \in B_{x^\star}(2\delta_{\rm ss})$ for every local-phase iterate and $q \leq 1+p$. Then*

$$K_{\rm loc}^H \;\leq\; \left\lceil \frac{L_1\delta_{\rm ss}^2}{2c_{\mathcal{N}}} \right\rceil. \tag{5.9}$$

Proof. By (3.7) and (5.7), a step with $\|g_k\| \geq 1$ satisfies $\Delta_k - \Delta_{k+1} \geq c_{\mathcal{N}}\|g_k\|^{\min(a_p,b_p)}$. Since $q \leq 1+p$ gives $b_p \geq 0$, we have $\min(a_p,b_p) \geq 0$ and hence $\|g_k\|^{\min(a_p,b_p)} \geq 1$: each such step decreases Δ by at least $c_{\mathcal{N}}$. Summing over these steps, and using that every other local-phase step also decreases Δ together with the entry bound (5.3),

$$\frac{L_1\delta_{\rm ss}^2}{2} \;\geq\; \Delta_{k_0} \;\geq\; \sum_{k\,:\,\|g_k\|\geq 1} (\Delta_k - \Delta_{k+1}) \;\geq\; K_{\rm loc}^H\, c_{\mathcal{N}}.$$

□

The restriction $q \leq 1+p$ is the standing assumption of Theorem 1, so it costs nothing here. Outside it the same argument applies with $\|g_k\|^{b_p} \geq (L_1\delta_{\rm ss})^{b_p}$ from (5.4) in place of $\|g_k\|^{\min(a_p,b_p)} \geq 1$, at the price of a factor $(L_1\delta_{\rm ss})^{2(q-1-p)}$.

5.3.3 Small-gradient steps

For a step with $\|g_k\| \leq 1$, (3.7), (5.7) and the Polyak–Łojasiewicz inequality of (5.2) give

$$\Delta_k - \Delta_{k+1} \;\geq\; c_{\mathcal{N}}\,\|g_k\|^{2r} \;\geq\; c_{\mathcal{N}}\,(2\gamma_{\rm ss}\Delta_k)^r,$$

that is,

$$\Delta_{k+1} \;\leq\; \left(1 - c_{\mathcal{N}}(2\gamma_{\rm ss})^r\Delta_k^{r-1}\right)\Delta_k \qquad \text{when } \|g_k\| \leq 1. \tag{5.10}$$

The count $K_{\rm loc}^L$ splits into three subcases according to r. Throughout, k_1 denotes the first iteration with $\|g_k\| \leq 1$; by monotonicity, $\Delta_{k_1} \leq \Delta_{k_0} \leq L_1\delta_{\rm ss}^2/2$.

Subcase L1: $r = 1$ (linear convergence). This corresponds to $p = 1$ and $q \geq 1$, or $p < 1$ and $q = p$.

Lemma 21 (Subcase L1)**.** *Suppose $x_k \in B_{x^\star}(2\delta_{\rm ss})$ for every local-phase iterate, $r = 1$, and $2c_{\mathcal{N}}\gamma_{\rm ss} < 1$. Then*

$$K_{\rm loc}^L \;\leq\; \left\lceil \frac{1}{2c_{\mathcal{N}}\gamma_{\rm ss}}\,\log\left(\frac{L_1^2\delta_{\rm ss}^2}{\varepsilon_{\rm loc}^2}\right)\right\rceil. \tag{5.11}$$

Proof. Write $\theta_L \coloneqq 1 - 2c_{\mathcal{N}}\gamma_{\rm ss} \in (0,1)$ and let $k_1 \leq \tau_1 < \tau_2 < \dots$ be the steps with $\|g_k\| \leq 1$. With $r = 1$, (5.10) gives $\Delta_{\tau_i+1} \leq \theta_L\Delta_{\tau_i}$, and every other local-phase step decreases Δ_k, so

$$\Delta_{\tau_K+1} \;\leq\; \theta_L^K\,\Delta_{k_1}.$$

Suppose the stopping condition has not been met at step $\tau_K + 1$, that is $\Delta_{\tau_K+1} > \varepsilon_{\mathrm{loc}}^2/(2L_1)$. Taking logarithms and dividing by $\log \theta_L < 0$,

$$K \;<\; \frac{1}{-\log \theta_L} \log\!\left(\frac{2L_1 \Delta_{k_1}}{\varepsilon_{\mathrm{loc}}^2}\right) \;\le\; \frac{1}{2c_{\mathcal{N}}\gamma_{\mathrm{ss}}} \log\!\left(\frac{L_1^2 \delta_{\mathrm{ss}}^2}{\varepsilon_{\mathrm{loc}}^2}\right),$$

using $-\log(1-x) \ge x$ on $[0,1)$ for the second inequality and $\Delta_{k_1} \le L_1 \delta_{\mathrm{ss}}^2/2$ for the third. Hence the number of such steps is at most the ceiling of the right-hand side. □

Subcase L2: $r < 1$ (finitely many non-restart steps). This corresponds to $p < 1$ and $q > p$.

Lemma 22 (Subcase L2)**.** *Suppose $x_k \in B_{x^\star}(2\delta_{\mathrm{ss}})$ for every local-phase iterate and $r < 1$. Then*

$$K_{\mathrm{loc}}^L \;\le\; \left\lceil \frac{(L_1 \delta_{\mathrm{ss}}^2/2)^{1-r}}{(1-r)\, c_{\mathcal{N}}\, (2\gamma_{\mathrm{ss}})^r} \right\rceil. \tag{5.12}$$

Proof. Write $c_k := c_{\mathcal{N}}(2\gamma_{\mathrm{ss}})^r \Delta_k^{r-1}$, so that (5.10) reads $\Delta_{k+1} \le (1-c_k)\Delta_k$.

If $c_k \ge 1$ at some such step, then $\Delta_{k+1} \le 0$, and since $\Delta_{k+1} \ge 0$ we get $\Delta_{k+1} = 0$: the target is met and no further such step occurs. We may therefore assume $c_k \in (0,1)$ at every step counted below.

Define $\varphi_k := \Delta_k^{1-r}$. Since $1-r \in (0,1)$, the map $t \mapsto t^{1-r}$ is increasing in t on $(0,\infty)$, so raising $\Delta_{k+1} \le (1-c_k)\Delta_k$ to the power $1-r$ preserves the inequality, and the concavity bound $(1-c)^{1-r} \le 1-(1-r)c$ for $c \in (0,1)$ gives

$$\varphi_{k+1} \;\le\; \bigl(1-(1-r)c_k\bigr)\varphi_k \;=\; \varphi_k - (1-r)\, c_{\mathcal{N}}\, (2\gamma_{\mathrm{ss}})^r,$$

where the equality uses $c_k \varphi_k = c_{\mathcal{N}}(2\gamma_{\mathrm{ss}})^r \Delta_k^{r-1}\Delta_k^{1-r} = c_{\mathcal{N}}(2\gamma_{\mathrm{ss}})^r$.

Let $k_1 \le \tau_1 < \tau_2 < \dots$ denote the steps with $\|g_k\| \le 1$. On every other local-phase step Δ_k decreases, and $1-r > 0$, so $\varphi_k = \Delta_k^{1-r}$ does not increase there either. Telescoping over this subsequence therefore gives

$$\varphi_{\tau_K+1} \;\le\; \varphi_{k_1} - (1-r)\, c_{\mathcal{N}}\, (2\gamma_{\mathrm{ss}})^r K,$$

and since $\varphi_{\tau_K+1} \ge 0$,

$$K_{\mathrm{loc}}^L \;\le\; \frac{\varphi_{k_1}}{(1-r)\, c_{\mathcal{N}}\, (2\gamma_{\mathrm{ss}})^r} \;\le\; \frac{(L_1 \delta_{\mathrm{ss}}^2/2)^{1-r}}{(1-r)\, c_{\mathcal{N}}\, (2\gamma_{\mathrm{ss}})^r},$$

using $\varphi_{k_1} = \Delta_{k_1}^{1-r} \le (L_1 \delta_{\mathrm{ss}}^2/2)^{1-r}$. □

Remark 11 (Restart steps set the floor). For $r < 1$ the number of non-restart steps with $\|g_k\| \le 1$ is a constant independent of $\varepsilon_{\mathrm{loc}}$, but this does not make the local phase $O(1)$: by Lemma 3, for $q > p$ the safeguard is unsatisfiable below the gradient threshold $(\sigma/\kappa)^{1/(q-p)}$, so near $x^\star$ *every* step is a restart, and the local complexity is governed by the linear rate of Lemma 19: $O(\log(1/\varepsilon_{\mathrm{loc}}))$. The threshold lemma thus explains structurally why the restart count, not the non-restart count, sets the floor in this regime.

Subcase L3: $r > 1$ (sublinear convergence). This corresponds to $p > 1$ or $q < p$.

Lemma 23 (Subcase L3)**.** *Suppose $x_k \in B_{x^\star}(2\delta_{\mathrm{ss}})$ for every local-phase iterate, $r > 1$, and $c_{\mathcal{N}}(2\gamma_{\mathrm{ss}})^r \Delta_{k_1}^{r-1} < 1$. Then*

$$K_{\mathrm{loc}}^L \;\le\; \left\lceil \frac{(2L_1)^{r-1}}{(r-1)\, c_{\mathcal{N}}\, (2\gamma_{\mathrm{ss}})^r\, \varepsilon_{\mathrm{loc}}^{2(r-1)}} \right\rceil. \tag{5.13}$$

Proof. Write $c_k := c_{\mathcal{N}}(2\gamma_{\rm ss})^r \Delta_k^{r-1}$. Since $r-1>0$ and Δ_k is nonincreasing, $c_k \le c_{k_1} < 1$ by hypothesis at every such step.

Define $\varphi_k := \Delta_k^{1-r}$; since $1-r<0$, the map $t \mapsto t^{1-r}$ is decreasing in t on $(0,\infty)$, so raising $\Delta_{k+1} \le (1-c_k)\Delta_k$ to the power $1-r$ reverses the inequality, and the convexity bound $(1-c)^{1-r} \ge 1-(1-r)c$ for $c \in (0,1)$ gives

$$\varphi_{k+1} \;\ge\; \big(1-(1-r)c_k\big)\varphi_k \;=\; \varphi_k + (r-1)\, c_{\mathcal{N}}\, (2\gamma_{\rm ss})^r,$$

using $c_k\varphi_k = c_{\mathcal{N}}(2\gamma_{\rm ss})^r$ as before.

Let $k_1 \le \tau_1 < \tau_2 < \dots$ denote the steps with $\|g_k\| \le 1$. On every other local-phase step Δ_k decreases, and $1-r<0$, so $\varphi_k = \Delta_k^{1-r}$ does not decrease there either. Telescoping over this subsequence gives

$$\varphi_{\tau_K+1} \;\ge\; \varphi_{k_1} + (r-1)\, c_{\mathcal{N}}\, (2\gamma_{\rm ss})^r K.$$

Since $1-r<0$, the stopping condition $\Delta_k \le \varepsilon_{\rm loc}^2/(2L_1)$ is equivalent to $\varphi_k \ge (2L_1/\varepsilon_{\rm loc}^2)^{r-1}$, so it is met at step τ_K+1 as soon as

$$\varphi_{k_1} + (r-1)\, c_{\mathcal{N}}\, (2\gamma_{\rm ss})^r\, K \;\ge\; \left(\frac{2L_1}{\varepsilon_{\rm loc}^2}\right)^{r-1}.$$

Dropping $\varphi_{k_1} \ge 0$ yields (5.13). $\square$

5.4 Total complexity

The local-phase iteration count splits into restart and non-restart contributions, the latter split according to whether $\|g_k\| \ge 1$ or $\|g_k\| \le 1$:

$$K_{\rm loc} \;=\; K_{\rm loc}^{\mathcal{R}} + K_{\rm loc}^{H} + K_{\rm loc}^{L}.$$

Theorem 2 (Local phase complexity)**.** *Let f be $(\zeta_{\rm ss},\beta_{\rm ss},\gamma_{\rm ss},\delta_{\rm ss})$-strict saddle and let $\varepsilon_{\rm loc} \in (0,\varepsilon_g)$. Run Algorithm 2 with $\varepsilon_g < \zeta_{\rm ss}$ and $\varepsilon_H < \beta_{\rm ss}$, and suppose that*

(i) $q \le 1+p$;

(ii) the local-phase iterates remain in $B_{x^\star}(2\delta_{\rm ss})$;

(iii) $2\gamma_{\rm ss}c_{\mathcal{R}} < 1$; and, where applicable, $2c_{\mathcal{N}}\gamma_{\rm ss} < 1$ (case $r=1$) and $c_{\mathcal{N}}(2\gamma_{\rm ss})^r(L_1\delta_{\rm ss}^2/2)^{r-1} < 1$ (case $r>1$), the latter implying the condition of Lemma 23 at Δ_{k_1} by (5.3).

Then, on the success event of Theorem 1, the local phase reaches $\Delta_k \le \varepsilon_{\rm loc}^2/(2L_1)$, and hence $\|\nabla f(x_k)\| \le \varepsilon_{\rm loc}$, within

$$K_{\rm loc} \;\le\; \underbrace{\left\lceil \frac{1}{2\gamma_{\rm ss}c_{\mathcal{R}}}\,\log\left(\frac{L_1^2\delta_{\rm ss}^2}{\varepsilon_{\rm loc}^2}\right)\right\rceil}_{K_{\rm loc}^{\mathcal{R}}} + \underbrace{\left\lceil \frac{L_1\delta_{\rm ss}^2}{2c_{\mathcal{N}}}\right\rceil}_{K_{\rm loc}^{H}} + \underbrace{\begin{cases} \left\lceil \dfrac{1}{2c_{\mathcal{N}}\gamma_{\rm ss}}\,\log\left(\dfrac{L_1^2\delta_{\rm ss}^2}{\varepsilon_{\rm loc}^2}\right)\right\rceil & \text{if } r=1,\\[2ex] \left\lceil \dfrac{(L_1\delta_{\rm ss}^2/2)^{1-r}}{(1-r)\, c_{\mathcal{N}}\, (2\gamma_{\rm ss})^r}\right\rceil & \text{if } r<1,\\[2ex] \left\lceil \dfrac{(2L_1)^{r-1}}{(r-1)\, c_{\mathcal{N}}\, (2\gamma_{\rm ss})^r\, \varepsilon_{\rm loc}^{2(r-1)}}\right\rceil & \text{if } r>1, \end{cases}}_{K_{\rm loc}^{L}} \quad (5.14)$$

where r is as in (5.5).

Proof. Every local-phase iteration is a restart step or a non-restart step, and every non-restart step has $\|g_k\| \ge 1$ or $\|g_k\| \le 1$ by (5.7). The three counts are bounded by Lemma 19, by Lemma 20, valid under hypothesis (i), and by whichever of Lemmas 21 to 23 applies according to r. Summing gives (5.14). $\square$

Remark 12 (The confinement hypothesis). Hypothesis (ii) assumes that the local-phase iterates stay in $B_{x^\star}(2\delta_{\mathrm{ss}})$, where f is strongly convex. We do not know how to prove it. Lemma 18 puts x_{k_0} within δ_{ss} of $x^\star$ and every step decreases f, but neither bounds the length of a single step, so an iterate could leave the basin. A proof would need control of f along the search segments and not only at the iterates, which the Armijo rule started from $\alpha = 1$ does not provide. We leave this open.

Corollary 1 (Asymptotic local complexity)**.** *Under the hypotheses of Theorem 2, as $\varepsilon_{\mathrm{loc}} \to 0$ with the strict-saddle parameters $(\zeta_{\mathrm{ss}}, \beta_{\mathrm{ss}}, \gamma_{\mathrm{ss}}, \delta_{\mathrm{ss}})$ held fixed,*

$$K_{\mathrm{loc}} \;=\; \begin{cases} O\big(\log(1/\varepsilon_{\mathrm{loc}})\big) & \text{if } r \le 1,\\ O\big(\varepsilon_{\mathrm{loc}}^{-2(r-1)}\big) & \text{if } r > 1, \end{cases} \tag{5.15}$$

with constants depending on $L_1, \gamma_{\mathrm{ss}}, c_{\mathcal{R}}, c_{\mathcal{N}}, \delta_{\mathrm{ss}}$ and r but not on $\varepsilon_{\mathrm{loc}}$.

Remark 13 (When is the floor attained?). Remark 11 settles the case $q > p$. For $q = p$ the safeguard imposes no threshold on the gradient scale, and how often restarts occur near $x^\star$ depends on the algorithm and its parameters. This does not affect (5.15): in that regime restart and non-restart steps contract at comparable linear rates (Lemma 19 and Subcase L1), so the order of the bound is the same whichever type dominates.

Corollary 2 (End-to-end complexity of Algorithm 2)**.** *Let Assumptions 1 to 3 hold, let f be $(\zeta_{\mathrm{ss}}, \beta_{\mathrm{ss}}, \gamma_{\mathrm{ss}}, \delta_{\mathrm{ss}})$-strict saddle, and let the hypotheses of Theorems 1 and 2 hold with $\varepsilon_g < \zeta_{\mathrm{ss}}$, $\varepsilon_H < \beta_{\mathrm{ss}}$ and $\varepsilon_{\mathrm{loc}} \in (0, \varepsilon_g)$. Then, with probability at least $1 - \delta_{\mathrm{tot}}$, Algorithm 2 outputs a point x_k satisfying*

$$\|x_k - x^\star\| \;\le\; \frac{\varepsilon_{\mathrm{loc}}}{\sqrt{\gamma_{\mathrm{ss}} L_1}}$$

for some local minimizer $x^\star$ of f, after at most $N + K_{\mathrm{loc}}$ iterations, with N as in Theorem 1 and K_{loc} as in Theorem 2.

Proof. On the success event of Theorem 1, which has probability at least $1 - \delta_{\mathrm{tot}}$, Lemma 18 places the returned point inside the γ_{ss}-strongly convex basin, so the local phase starts there and Theorem 2 applies on the same event. It gives $\Delta_k \le \varepsilon_{\mathrm{loc}}^2/(2L_1)$ within K_{loc} iterations, and the first line of (5.2) then yields

$$\|x_k - x^\star\| \;\le\; \sqrt{\frac{2\Delta_k}{\gamma_{\mathrm{ss}}}} \;\le\; \sqrt{\frac{\varepsilon_{\mathrm{loc}}^2}{\gamma_{\mathrm{ss}} L_1}} \;=\; \frac{\varepsilon_{\mathrm{loc}}}{\sqrt{\gamma_{\mathrm{ss}} L_1}}.$$

The iteration count is the sum of the N iterations of Algorithm 1 and the K_{loc} iterations of the local phase. $\square$

Chapter 6

Conclusion

This work gives a second-order complexity guarantee for a method whose descent phase is a NCG iteration, a class for which no such guarantee was previously available. The method alternates a safeguarded restarted NCG phase with a perturbed HB escape phase, and returns an $(\varepsilon_g, \varepsilon_H)$-SOSP with high probability under Lipschitz gradient and Hessian alone. The escape phase uses gradient evaluations only, and replaces the negative-curvature oracle that hybrid methods of Section 2.3 invoke at every candidate stationary point.

The two obstructions identified in Section 2.7 shaped the design of Algorithm 1. NCG is still not analysed in isolation: it occupies the descent phase, where its restart safeguard supplies a per-step decrease. Inside the escape window the stepsize and momentum are constant and the safeguard is suspended, which is what makes the difference recursion (3.17) autonomous and its remainder proportional to the gap.

The worst-case bound does not improve on the state of the art. Under the single-tolerance coupling it matches perturbed gradient descent and falls short of the accelerated $\widetilde{O}(\varepsilon_g^{-7/4})$; Remark 9 explains why the constant momentum $\gamma < \frac{1}{2}$ that the localization energy requires cannot be pushed to the regime where acceleration would begin. The numerical experiments are more favourable, particularly when the budget is counted in oracle evaluations rather than iterations, but they rest on two planted instances with starts chosen on the stable manifold, and the gap between the certified and the executable constants (Section 4.2) is several orders of magnitude.

We believe any of the following directions would make interesting future work.

Confinement in the local phase. Theorem 2 assumes the local iterates remain in $B_{x^\star}(2\delta_{\mathrm{ss}})$. Lemma 18 places the first one within δ_{ss} of $x^\star$ and every step decreases f, but nothing bounds the length of a single step, so an iterate could leave the basin. A proof would need control of f along the search segments and not only at the iterates, which the Armijo rule started from $\alpha = 1$ does not provide (Remark 12).

A variable window. The coupling requires the two runs to share their stepsize and momentum, and $\mathbf{M}$ to be constant. Allowing a line search inside the window would require bounding $|\alpha_t - \alpha_t'|$ by the gap $\|w_t\|$, plausible for a line search Lipschitz in its starting point but not available from the Armijo rule as stated. A momentum schedule $\gamma_t \le \bar{\gamma} < \frac{1}{2}$ is a more tractable intermediate case: the energy of Section 3.3.1 extends without difficulty, and only the spectral analysis of Section 3.3.3, which bounds powers of a fixed matrix, would have to be redone for a product of distinct ones.

NCG in the window. Running restarted NCG in both phases would remove the phase split entirely. The obstruction is intrinsic rather than a matter of proof technique: two coupled runs carry different coefficients β_t, injecting a term $(\beta_t - \beta_t')d_{t-1}'$ of the order of the direction rather than of the gap, and the Fletcher–Reeves denominator is singular exactly where the perturbation is applied.

An accelerated escape with a certificate. Remark 9 locates the difficulty at $\gamma \to 1$, where the escape rate would become $1+\sqrt{a}$ but the energy (3.12) ceases to be monotone. Finding a restart statistic that is simultaneously a progress certificate and a localizer compatible with the coupling, in the spirit of [33], is the problem this architecture isolates.

Closing the ledger gap. The runs of Chapter 4 drop constraint (C2), whose constant Λ carries the accumulated worst-case slack of the induction of Lemma 12. Tightening that induction, and the logarithmic factor in (C4) that Section 4.5 shows is not decorative, would narrow the distance between what is certified and what can be executed.

Appendix A

Technical lemmas: powers of 2×2 matrices

The escape analysis of Section 3.3 reduces, through the block decomposition of Lemma 7, to bounding powers of 2×2 matrices at a prescribed geometric rate. We prove here the two elementary bounds this requires, according to whether the two eigenvalues are well separated or nearly equal. Throughout, $B \in \mathbb{C}^{2\times 2}$ has eigenvalues μ_1, μ_2 and spectral radius $\rho = \max\{|\mu_1|, |\mu_2|\}$.

The first bound covers the non-degenerate case. It is sharp when the eigenvalues are separated, but its constant scales with the inverse gap $|\mu_1 - \mu_2|^{-1}$ and is therefore unusable near a confluence.

Lemma 24 (Powers via Cayley–Hamilton)**.** *If $\mu_1 \neq \mu_2$, then for every $m \geq 1$,*

$$\|B^m\| \leq C(B)\,\rho^m, \qquad C(B) := \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|}. \tag{A.1}$$

The bound also holds at $m = 0$ whenever $C(B) \geq 1$.

Proof. Since B is 2×2, the Cayley–Hamilton theorem lets us write every power as an affine combination of B and I,

$$B^m = a_m B + b_m I,$$

and evaluating this identity on each eigenvalue gives the scalar system $\mu_i^m = a_m \mu_i + b_m$, $i = 1, 2$. As $\mu_1 \neq \mu_2$ it solves to

$$a_m = \frac{\mu_1^m - \mu_2^m}{\mu_1 - \mu_2}, \qquad b_m = \frac{\mu_1 \mu_2^m - \mu_2 \mu_1^m}{\mu_1 - \mu_2}.$$

Bounding numerators termwise, $|\mu_1^m - \mu_2^m| \leq 2\rho^m$ and $|\mu_1\mu_2^m - \mu_2\mu_1^m| \leq 2\rho^{m+1}$, so

$$\|B^m\| \leq |a_m|\|B\| + |b_m| \leq \frac{2\rho^m}{|\mu_1 - \mu_2|}\|B\| + \frac{2\rho^{m+1}}{|\mu_1 - \mu_2|} = \frac{2(\|B\| + \rho)}{|\mu_1 - \mu_2|}\,\rho^m.$$

At $m = 0$ the left-hand side is $\|I\| = 1$ and the right-hand side is $C(B)$, so the bound holds as soon as $C(B) \geq 1$. $\square$

The second bound removes the dependence on the gap, at the cost of a factor m. It relies on the Schur form rather than diagonalization, and so applies verbatim in the confluent case $\mu_1 = \mu_2$, where B may be defective.

Lemma 25 (Powers via the Schur form)**.** *Let $T = \begin{pmatrix} \mu_1 & s \\ 0 & \mu_2 \end{pmatrix}$. Then for every $m \geq 1$,*

$$T^m = \begin{pmatrix} \mu_1^m & c_m \\ 0 & \mu_2^m \end{pmatrix}, \qquad c_m = s \sum_{j=0}^{m-1} \mu_1^j \mu_2^{m-1-j}, \qquad |c_m| \leq |s|\, m\, \rho^{m-1}. \tag{A.2}$$

If moreover $B = QTQ^*$ is a Schur form of B with Q unitary, then $|s| \le \|B\|_F$ and, for every $m \ge 1$,

$$\|B^m\| \le \sqrt{2}\,\rho^m + \|B\|_F\, m\, \rho^{m-1}. \tag{A.3}$$

Proof. Power formula. As a product of upper-triangular matrices T^m is upper triangular, and its diagonal is (μ_1^m, μ_2^m); only the $(1,2)$ entry c_m requires computation. Reading the $(1,2)$ entry of $T^m = T\,T^{m-1}$ gives the recurrence

$$c_m = \mu_1 c_{m-1} + s\,\mu_2^{m-1}, \qquad c_0 = 0.$$

We check by induction that $c_m = s \sum_{j=0}^{m-1} \mu_1^j \mu_2^{m-1-j}$. It holds at $m = 1$, where $c_1 = s$. If it holds at $m-1$, then reindexing gives

$$\mu_1 c_{m-1} = s \sum_{j=1}^{m-1} \mu_1^j \mu_2^{m-1-j},$$

and adding $s\mu_2^{m-1}$, the $j = 0$ term, restores the full sum. Each of the m summands has modulus at most ρ^{m-1}, hence $|c_m| \le |s|\, m\, \rho^{m-1}$. When $\mu_1 = \mu_2 = \mu$ the sum collapses to $m\mu^{m-1}$, recovering the Jordan superdiagonal.

Norm bound. The Frobenius norm is unitarily invariant: $\|QAQ^*\|_F^2 = \operatorname{tr}(QA^*Q^*QAQ^*) = \operatorname{tr}(A^*A) = \|A\|_F^2$. Hence $\|T\|_F = \|B\|_F$, and since

$$\|T\|_F^2 = |\mu_1|^2 + |s|^2 + |\mu_2|^2 \ge |s|^2,$$

we get $|s| \le \|B\|_F$. The spectral norm is likewise unitarily invariant, so $\|B^m\| = \|T^m\|$. Bounding it by the Frobenius norm and using (A.2),

$$\|B^m\| \;\le\; \|T^m\|_F \;=\; \sqrt{|\mu_1|^{2m} + |\mu_2|^{2m} + |c_m|^2},$$

and by $|\mu_1|^{2m} + |\mu_2|^{2m} \le 2\rho^{2m}$, $|c_m| \le |s|\, m\, \rho^{m-1}$ and $\sqrt{a+b} \le \sqrt{a} + \sqrt{b}$,

$$\|B^m\| \;\le\; \sqrt{2}\,\rho^m + |s|\, m\, \rho^{m-1} \;\le\; \sqrt{2}\,\rho^m + \|B\|_F\, m\, \rho^{m-1}.$$

□

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